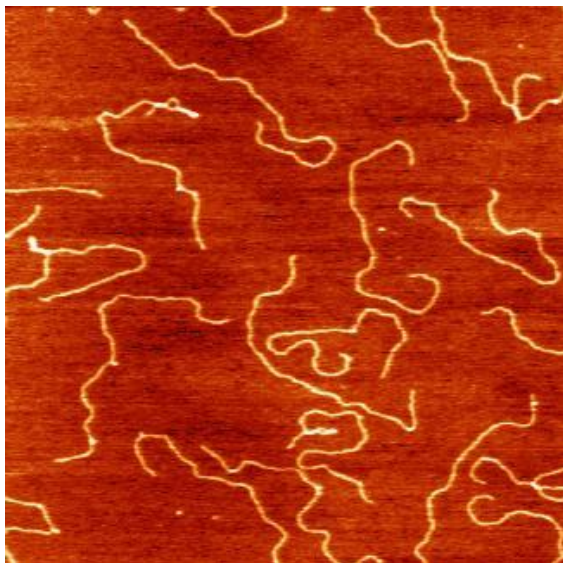


# 纳米测量技术概论

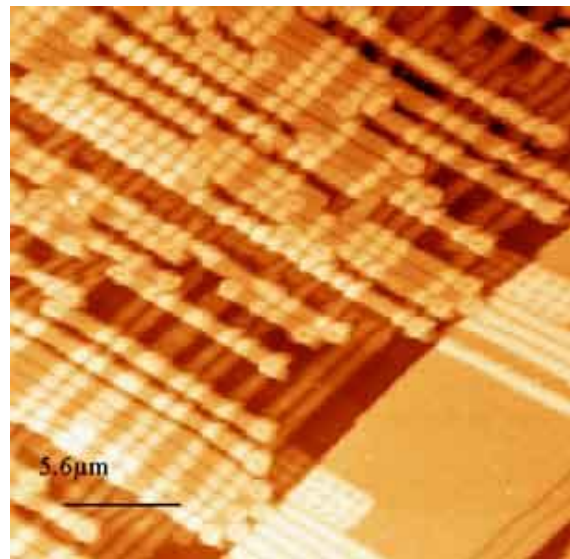
## 一 扫描探针显微术

2023-11-15

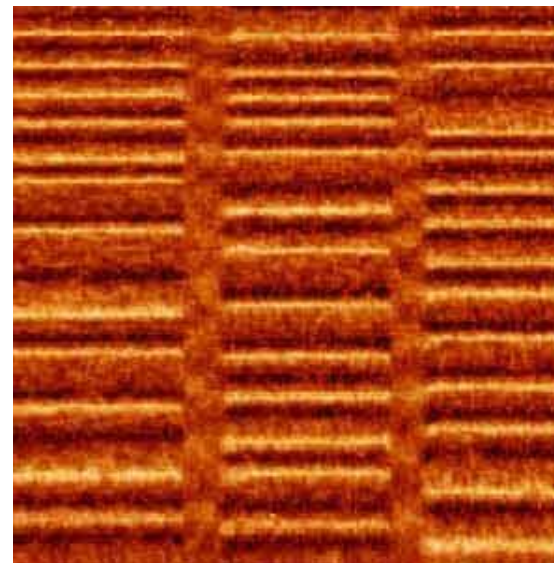
3A205



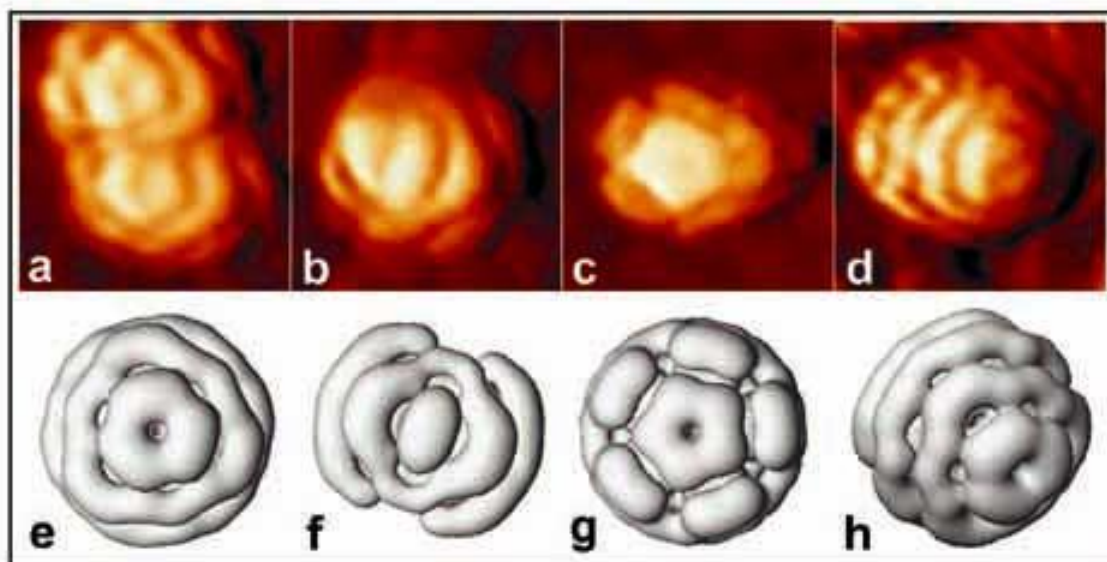
DNA



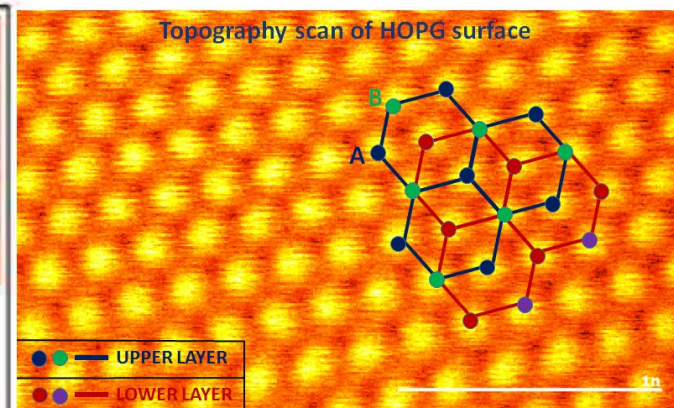
集成电路



磁盘



C60



石墨烯

# 纳米科技 -

1



Richard Feynman  
Cal Tech, 1959

1959年 R. Feynman演讲

2



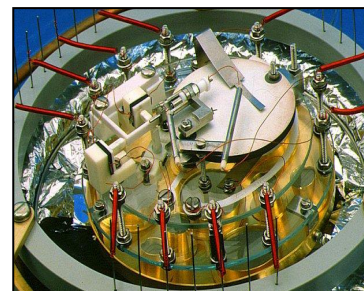
G. Binnig



H. Rohrer

1981年 STM发明

1986获诺贝尔物理奖

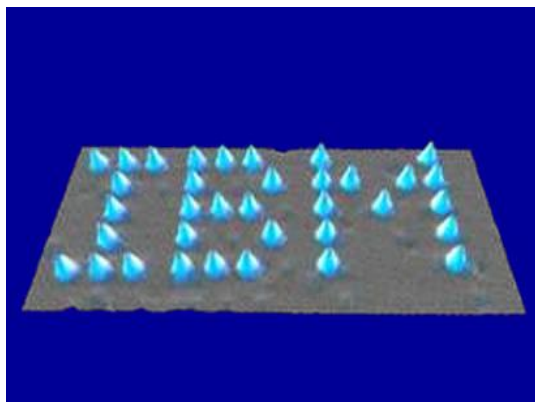


1981年 STM



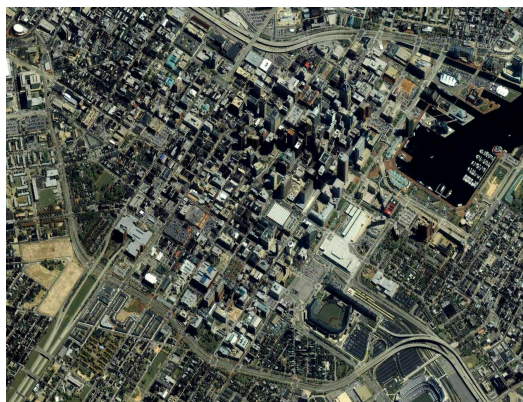
1986年 AFM

3

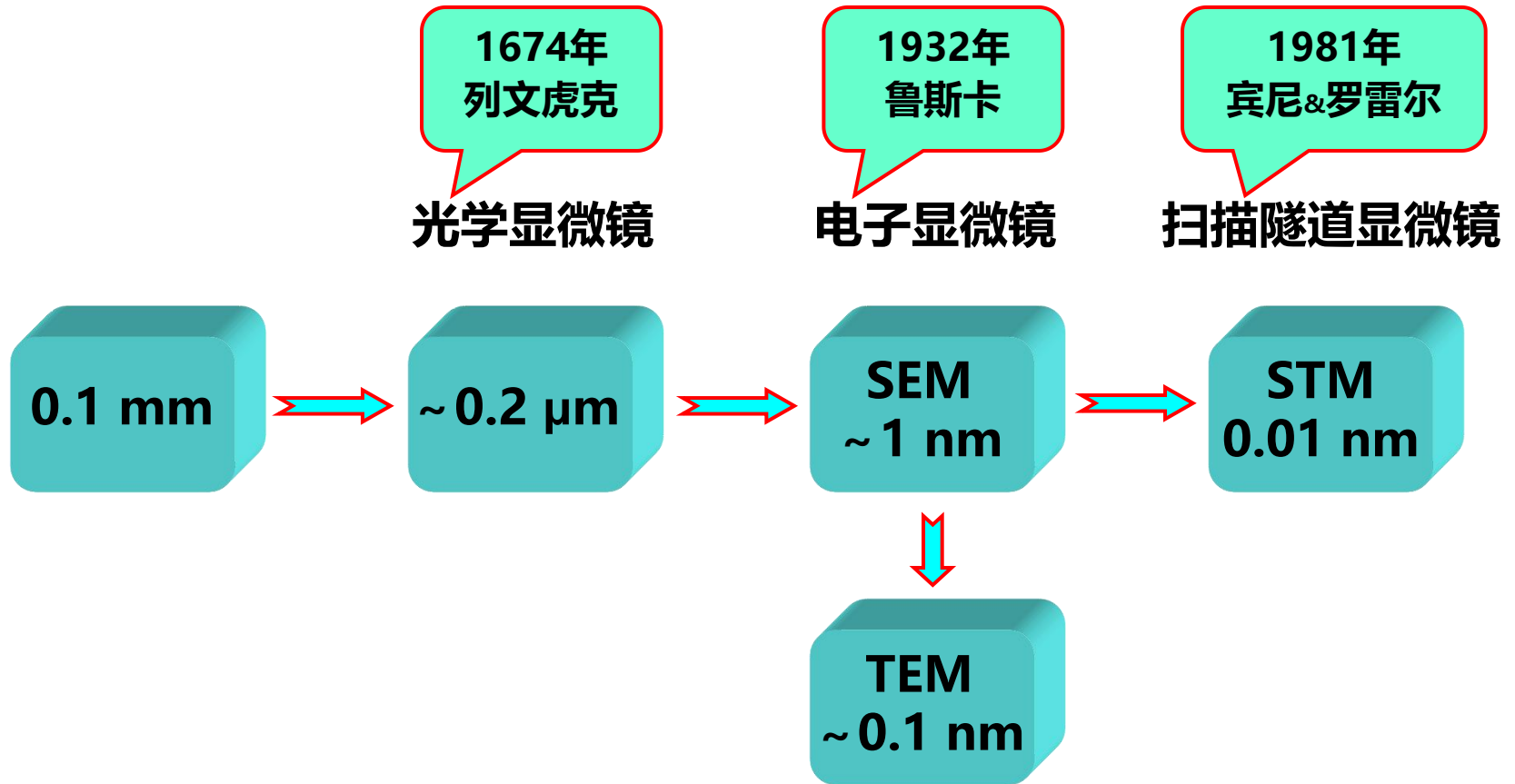


1989年 D. Eigler单原子操纵

4



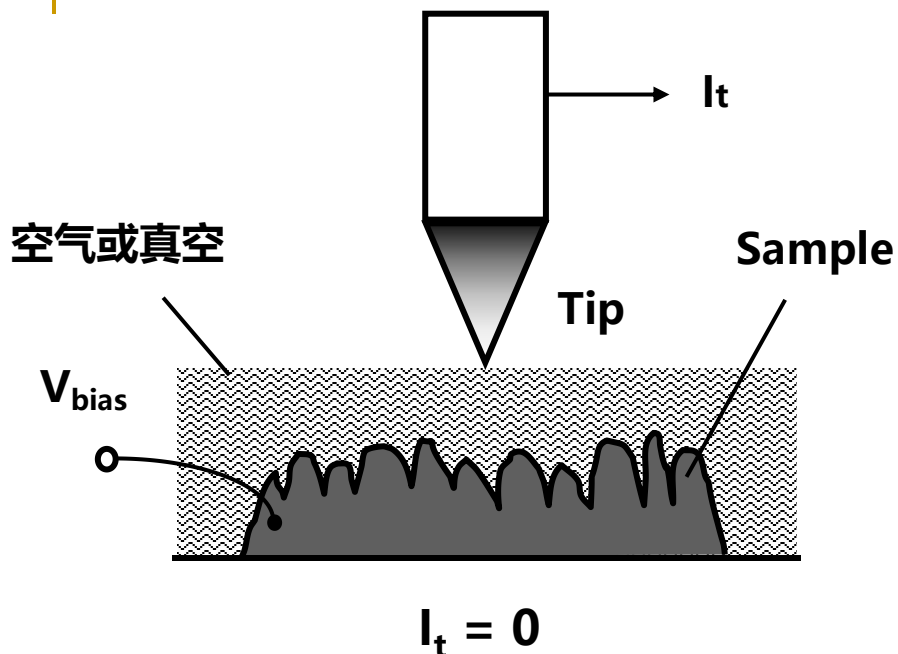
1990年 首届纳米科技  
国际会议 (巴尔的摩)



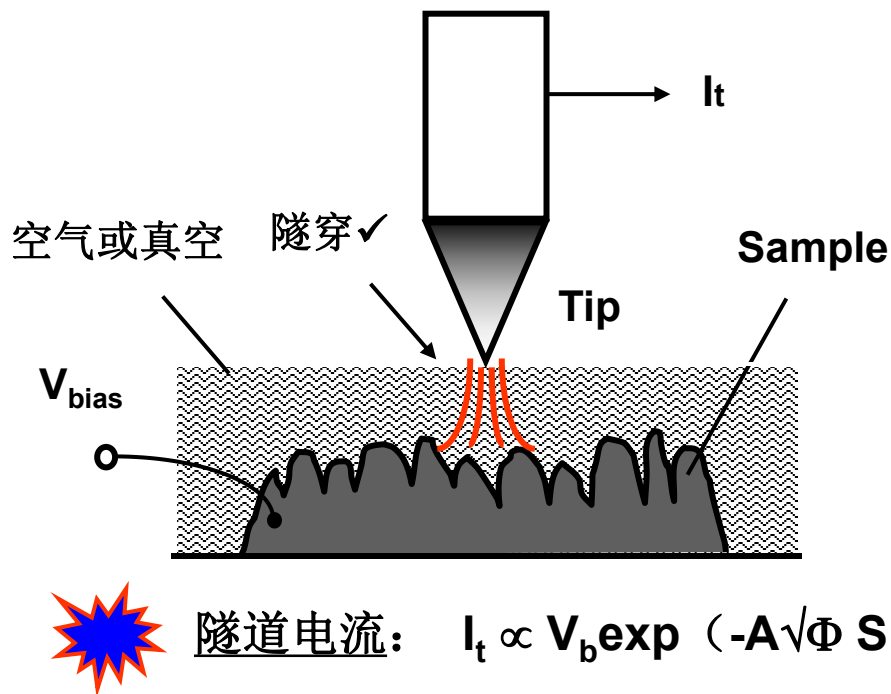
□ Shorter wavelengths → Electron microscopy (SEM, TEM)

□ Nearfield techniques (STM, AFM, SNOM, etc.)

## ✧ 经典物理学



## ✧ 量子力学—隧道效应

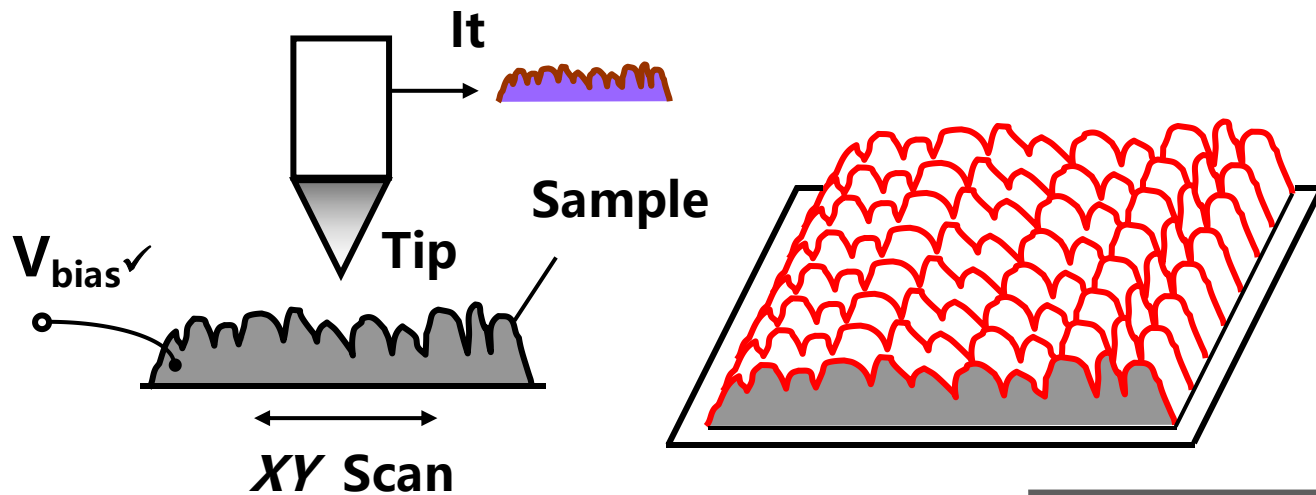


- A : 常数, 真空中  $\approx 1$
- $V_b$  : 偏压, 2 mV ~ 2 V
- $\phi$  : 表面功函数, 5 eV
- S : 间距, 0.3 nm ~ 1.0 nm

电焊?

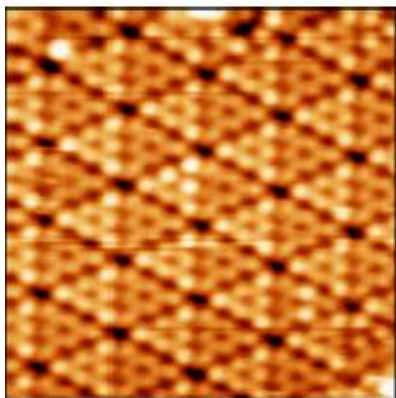
$$\Delta S \approx 0.1 \text{ nm} \Rightarrow \Delta I_t \approx 10 \text{ 倍! !}$$

# 扫描隧道显微镜原理

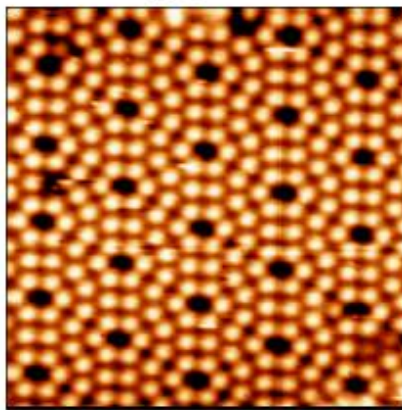


Si(111) 7x7

Filled States.



Empty States.



原子分辨率

实空间图像

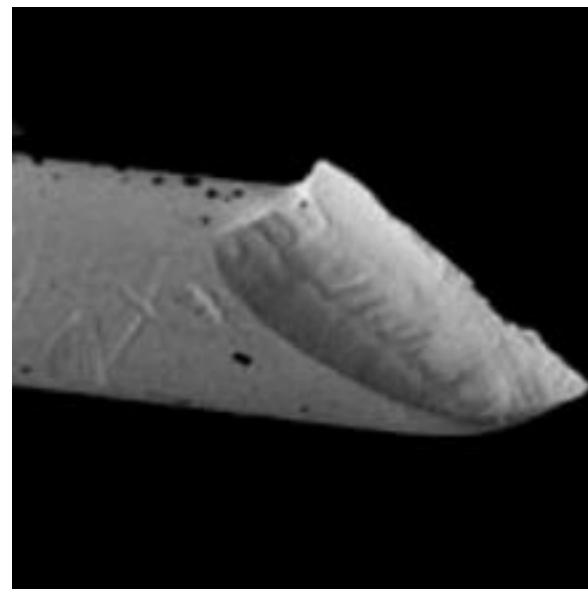
单原子层

全环境工作

隧道谱STS

## ✧ STM针尖

按材料分 { 钨丝针尖 (*tungsten*, *W*): 适用于真空  
铂—铱针尖 (*platinum-iridium*, *Pt-Ir*)



✧ Mechanical cutting

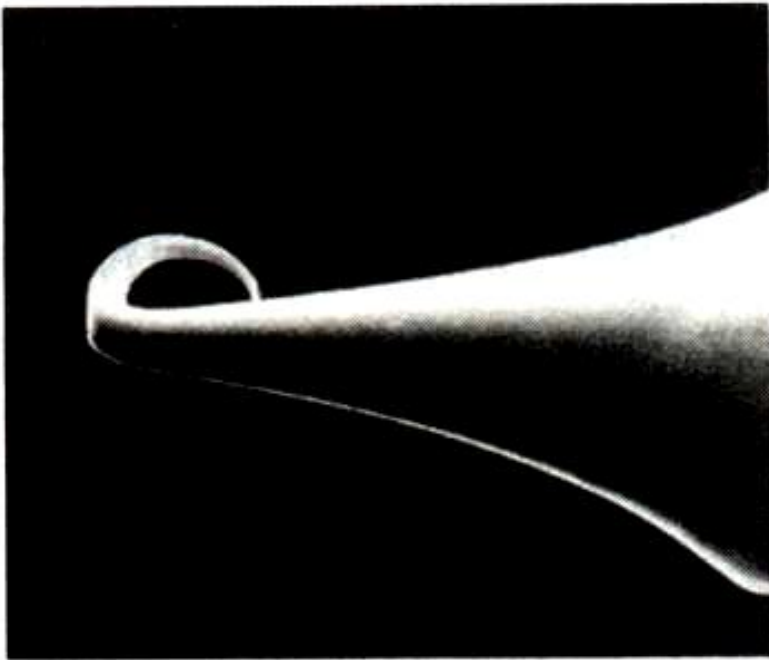
为什么可用“**粗大**”的探针探测出**极细小**的原子？

因为针尖总有一个原子凸出！

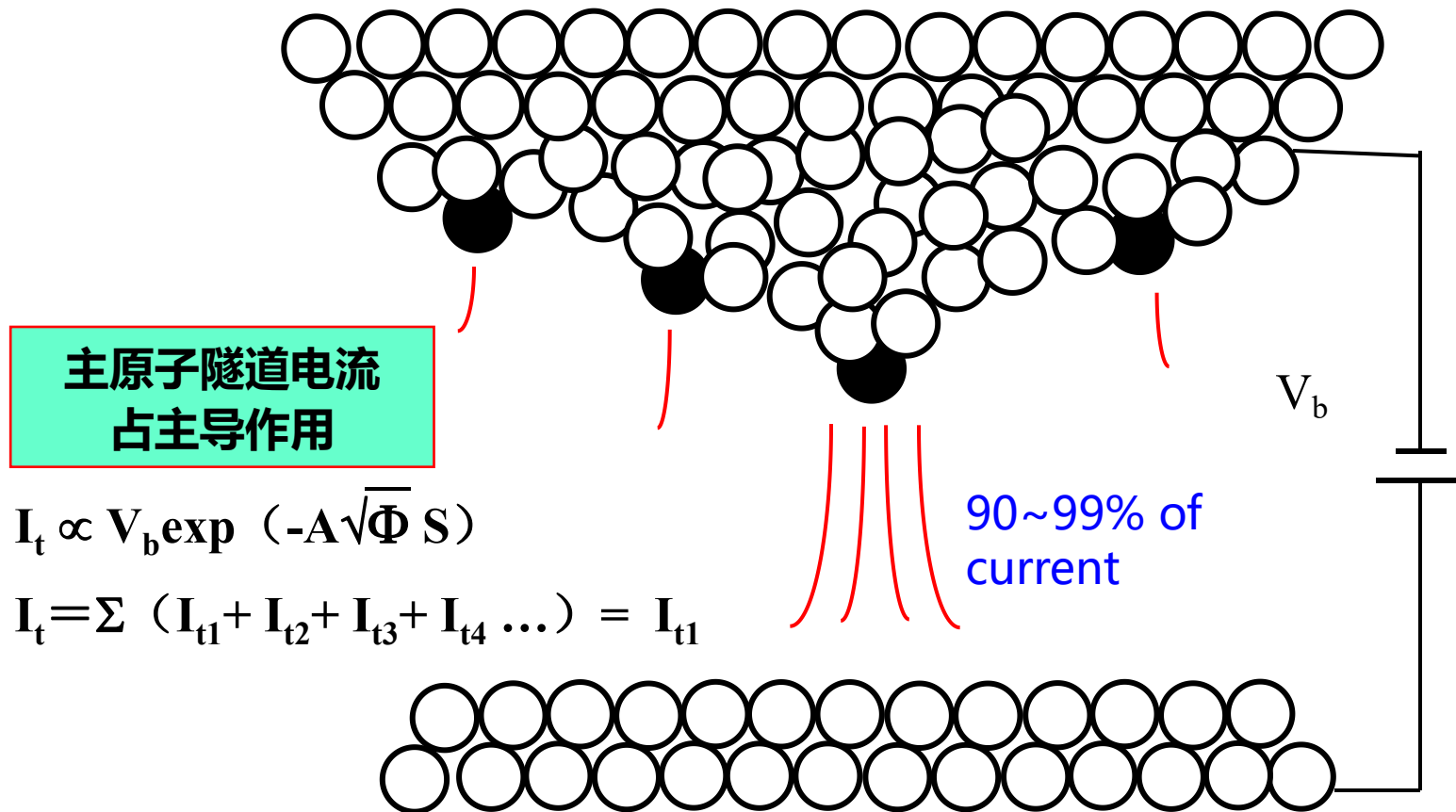
为什么可用“**粗大**”的手指头摸出细小的头发？

因为神经末梢比头发小！

Why

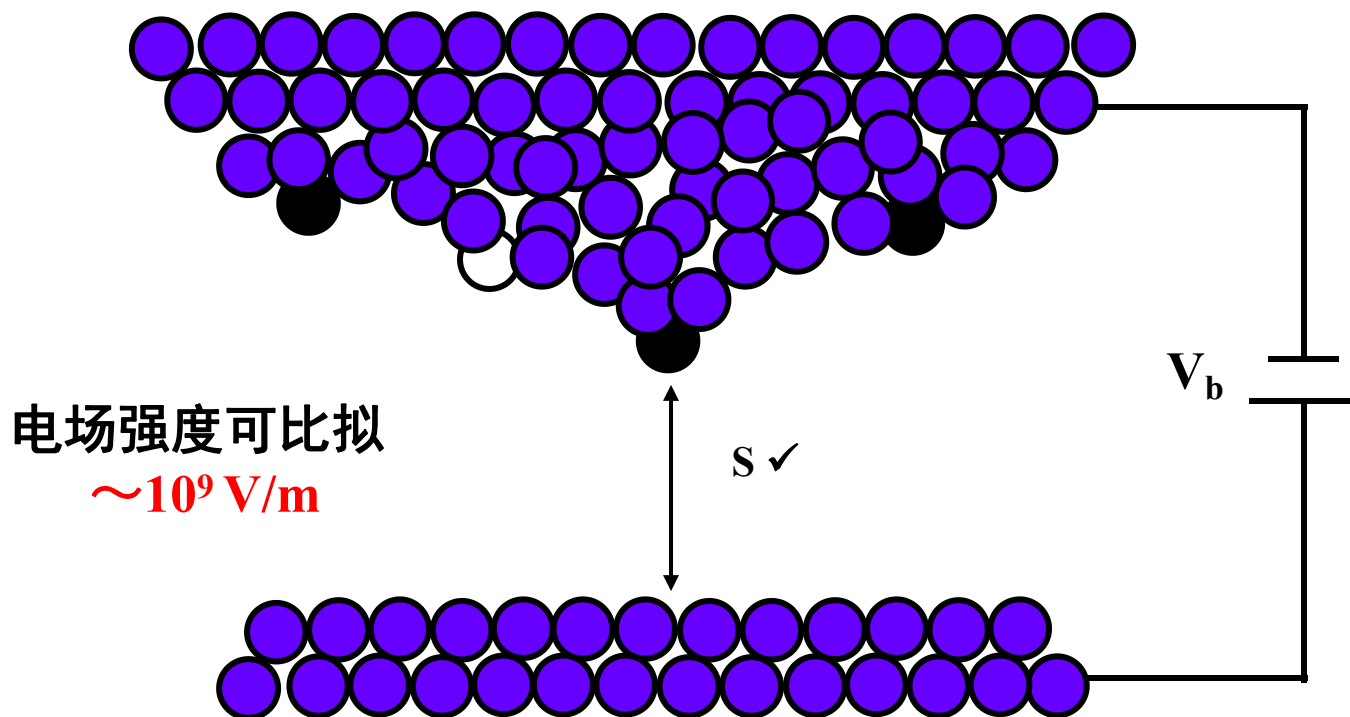


## 扫描测试原子图像时，隧道电流起主导作用



## 进行单原子操纵时，电场强度起主导作用

$$E \propto V_b / S, \quad \mathbf{E} \approx \mathbf{E}_1 \approx \mathbf{E}_2 \approx \mathbf{E}_3 \approx \mathbf{E}_4 \approx \dots$$

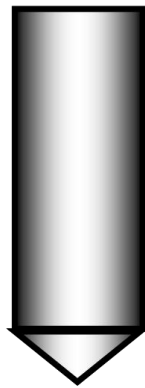


**Which tip is better or the best ?**

1



2



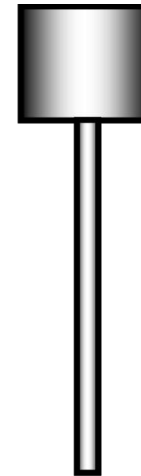
3



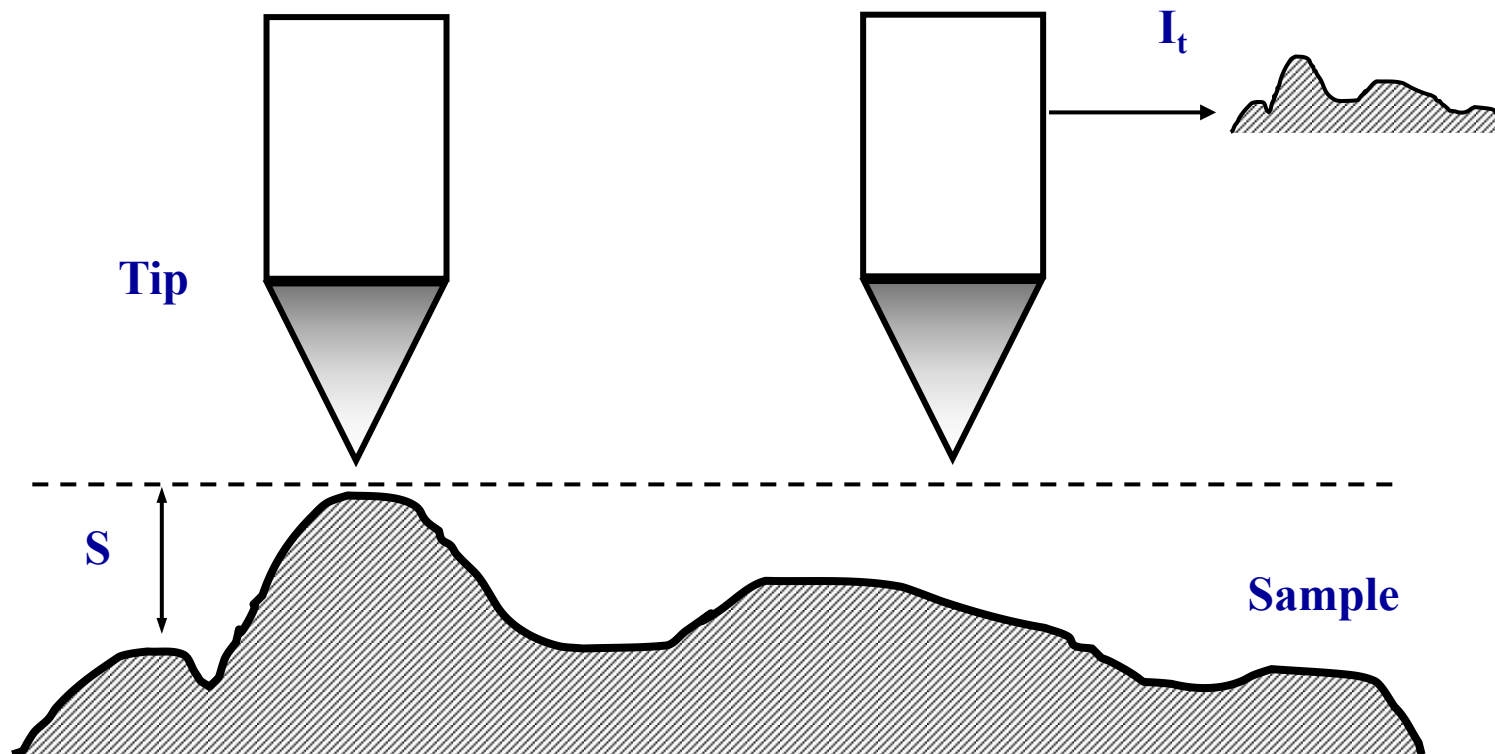
4



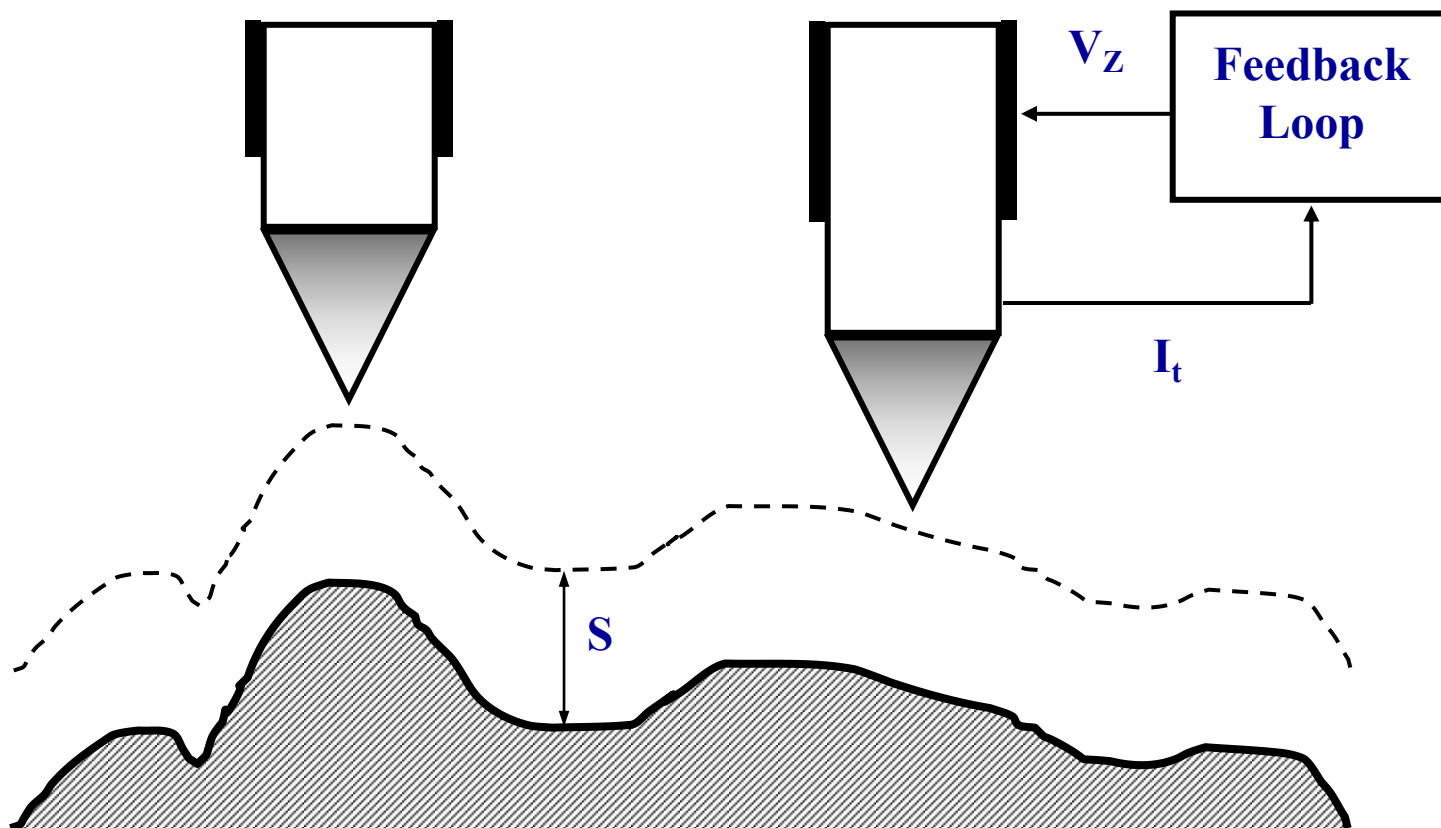
5



# 工作模式



等高模式



恒流模式

## STM两种扫描工作模式的性能比较

| 扫描模式 | 反 馈 | 特 征        | 扫描范围 | 分辨率 |
|------|-----|------------|------|-----|
| 等高模式 | 关 闭 | 针尖高度<br>恒定 | 较小   | 很高  |
| 恒流模式 | 开 启 | 隧道电流<br>恒定 | 较大   | 较高  |

# Characteristics of STM

**Very powerful tool for imaging and spectroscopy at atomic length scale**

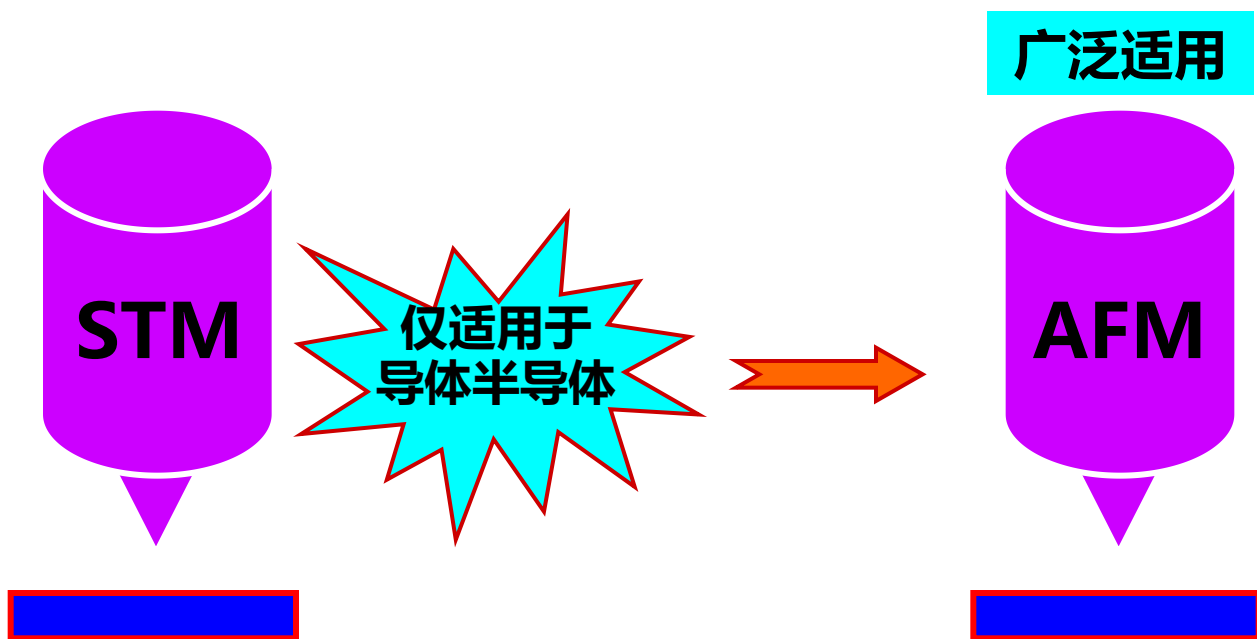
## **Advantages:**

- high spatial resolution (atomic resolution in best cases) in plane and out of plane
- possible to image local regions, defects, active sites, etc
- surface specific, surface states
- spatially resolved spectroscopy to get local information on electronic states
- local manipulation of sample (e.g. moving atoms, initiating local chemical reactions)

## **Disadvantages:**

- requires a conducting surface
- requires vacuum for highest-resolution operation (but can also operate in air or in solution)
- very sensitive to surface condition
- can be difficult to identify atomic structures unambiguously
- difficult to follow dynamic processes due to slow imaging

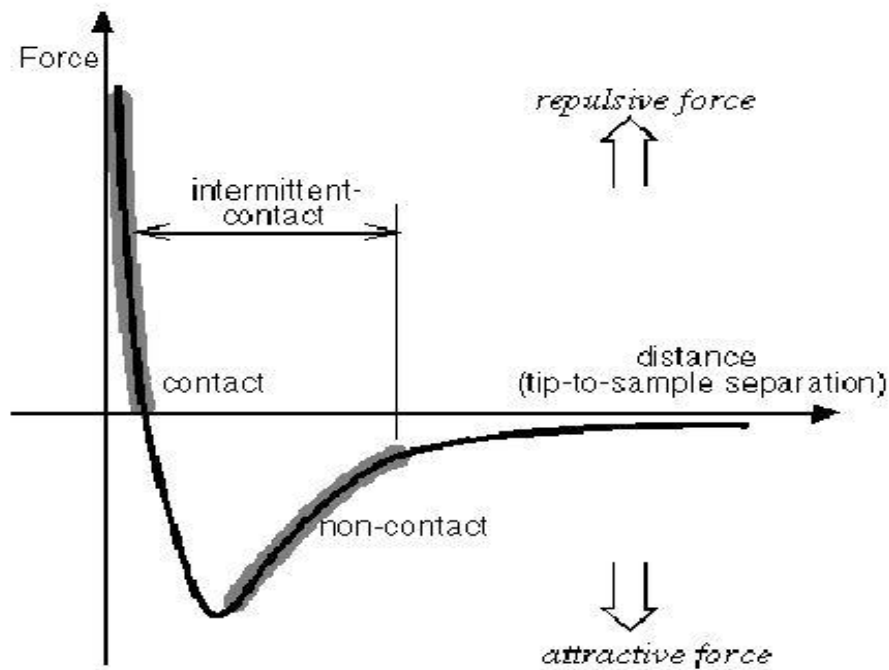
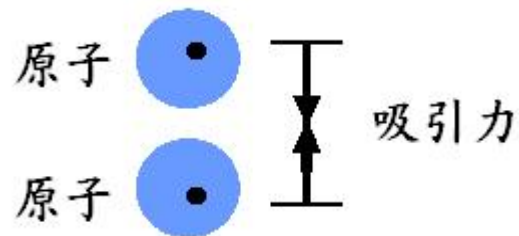
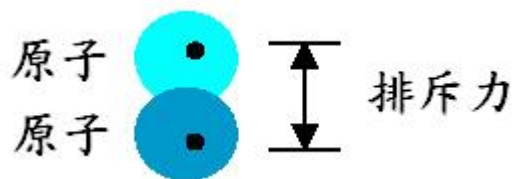
# 原子力显微镜



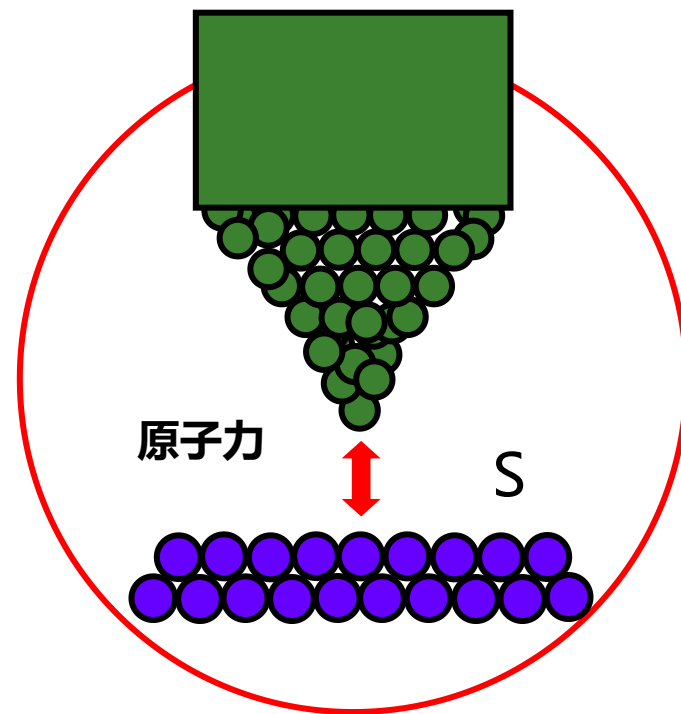
- STM : 电流

- AFM : 力

反馈用距离敏感物理量



Interatomic force vs. distance curve.



## 原子间作用力

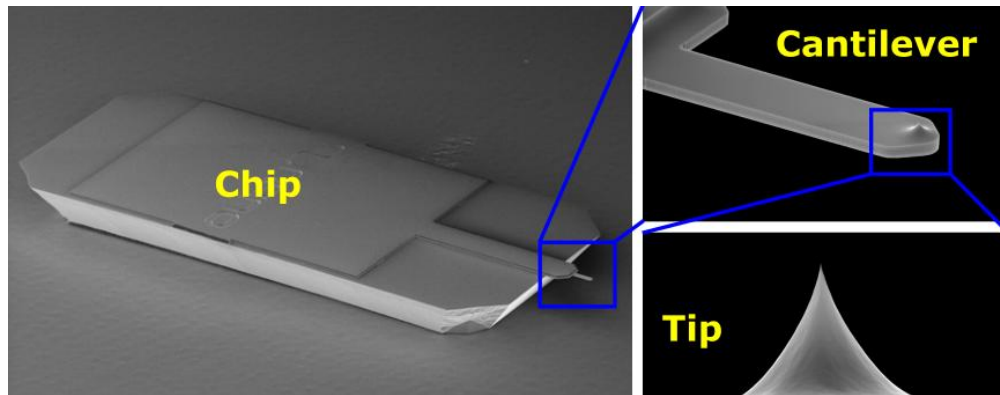
探针逼近样品时，两者间产生原子力：对应于探针/样品的间距。

# Estimate for Forces

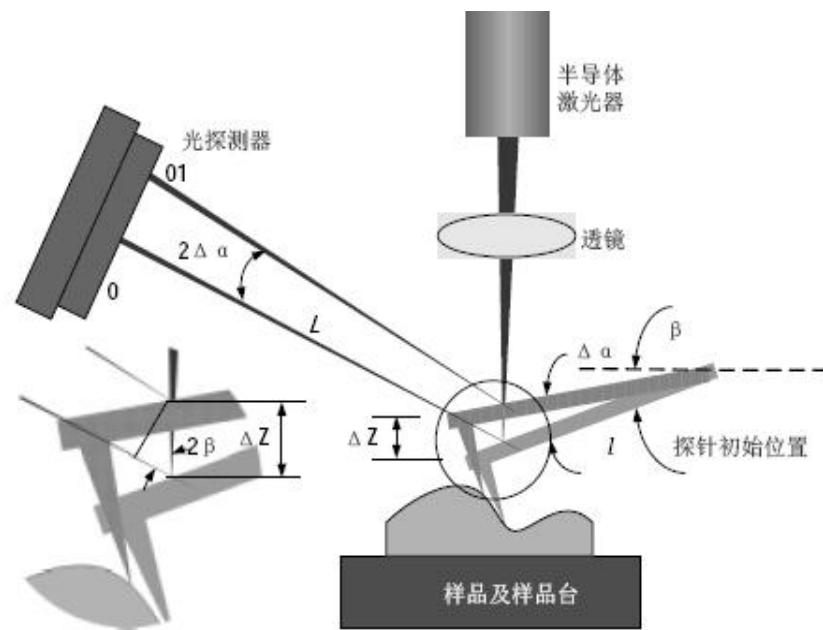
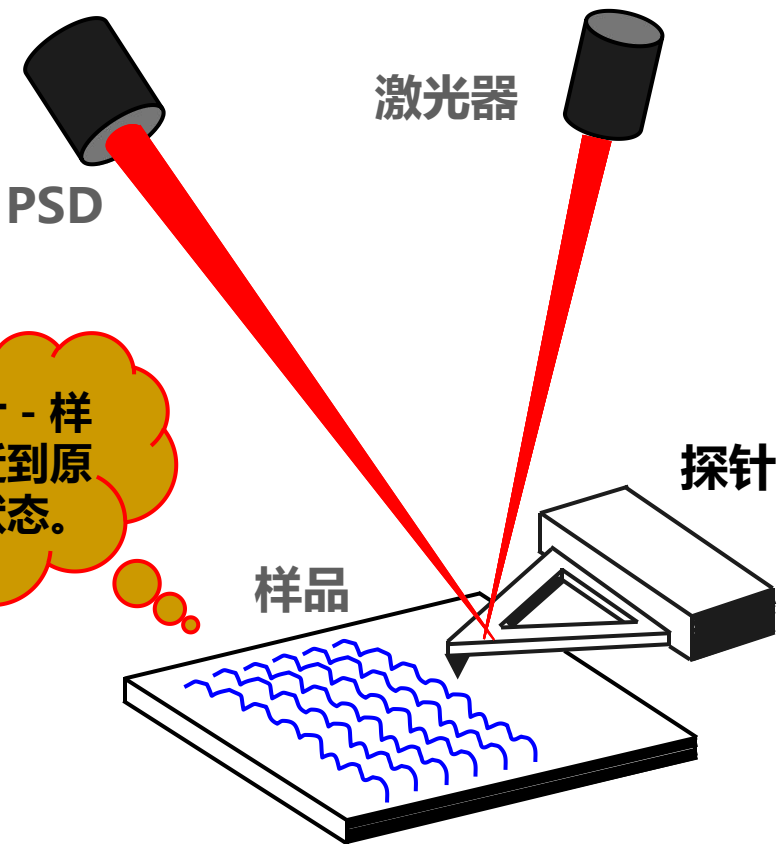
## 如何检测 微悬臂偏转量



因原子力引起的探针位移量**极小**



- 1. 隧道电流法
- 2. 电容检测法
- 3. 干涉法
- 4. 自感应法 (压阻、压电...)

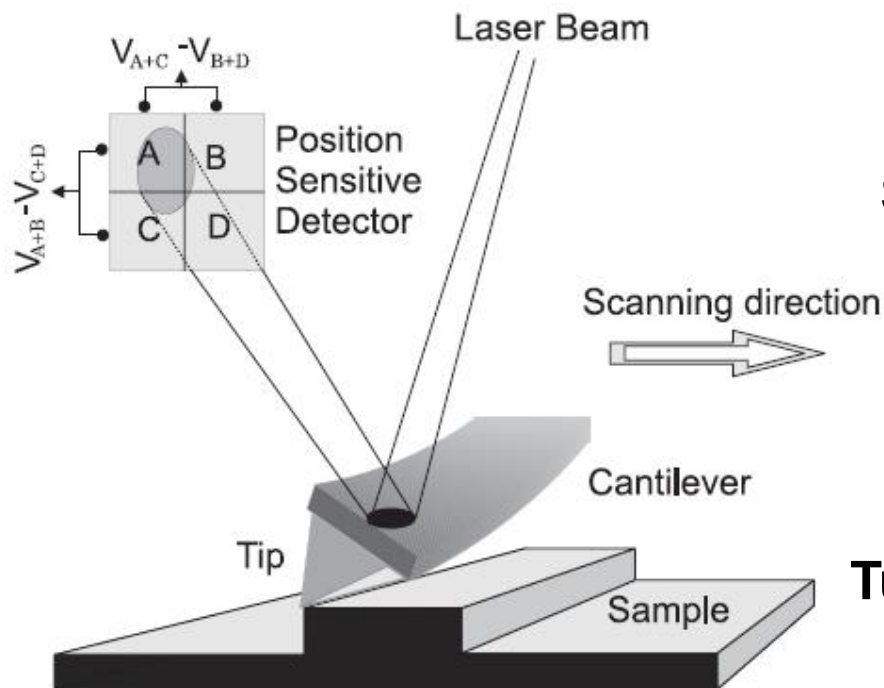


光点偏转法的微位移检测原理图

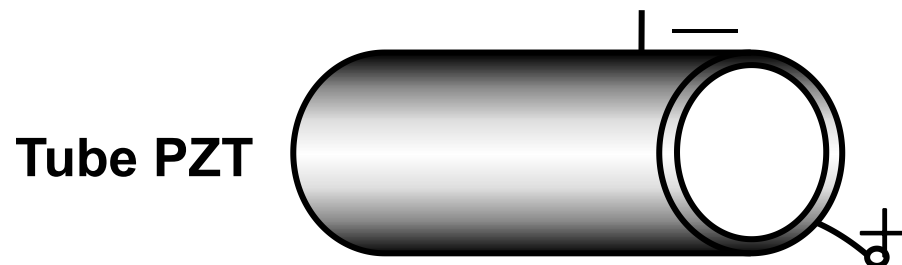
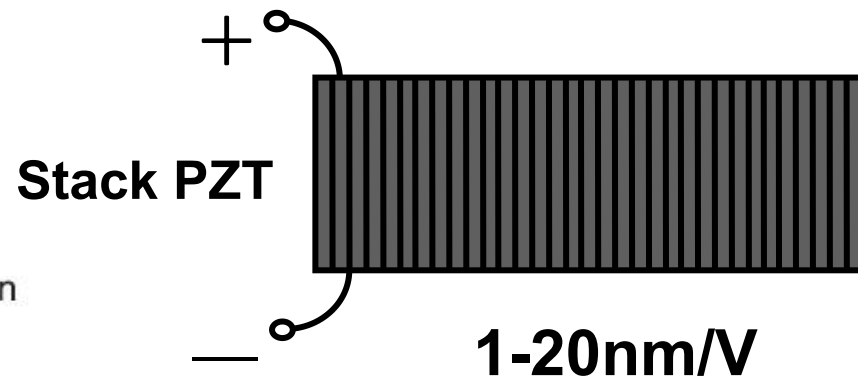
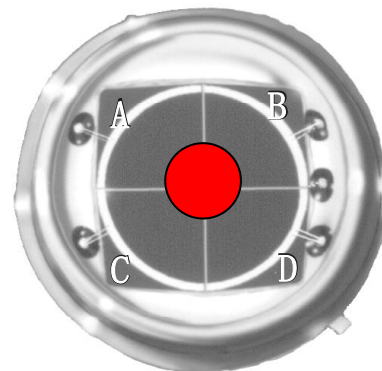
$$\Delta s \approx L \times 2\Delta\alpha = L \times 2 \times \frac{\Delta z}{l} = \frac{2L}{l} \times \Delta z$$

## 5. 光点偏转法

$$\begin{cases} X = \frac{k(I_A + I_C - I_B - I_D)}{I_A + I_B + I_C + I_D} \\ Y = \frac{k(I_A + I_B - I_C - I_D)}{I_A + I_B + I_C + I_D} \end{cases}$$

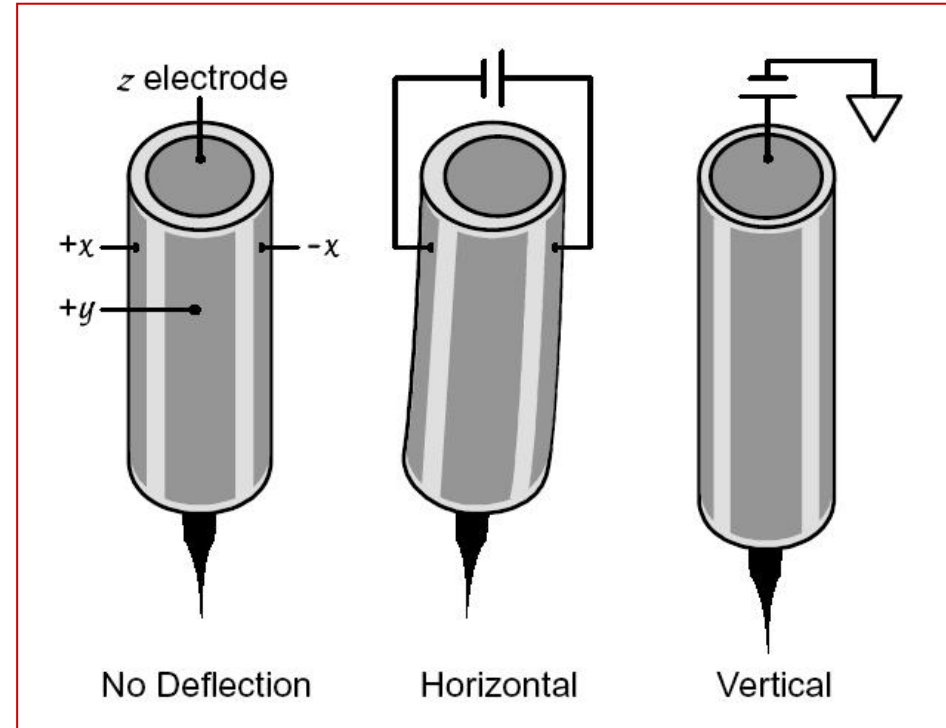
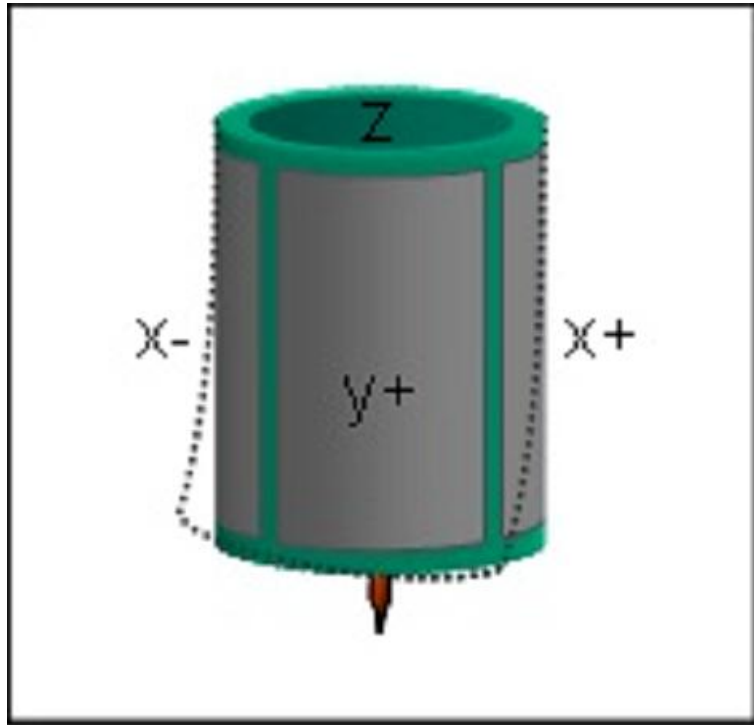


悬臂梁变形检测传感器



样品扫描台

# 单管扫描器



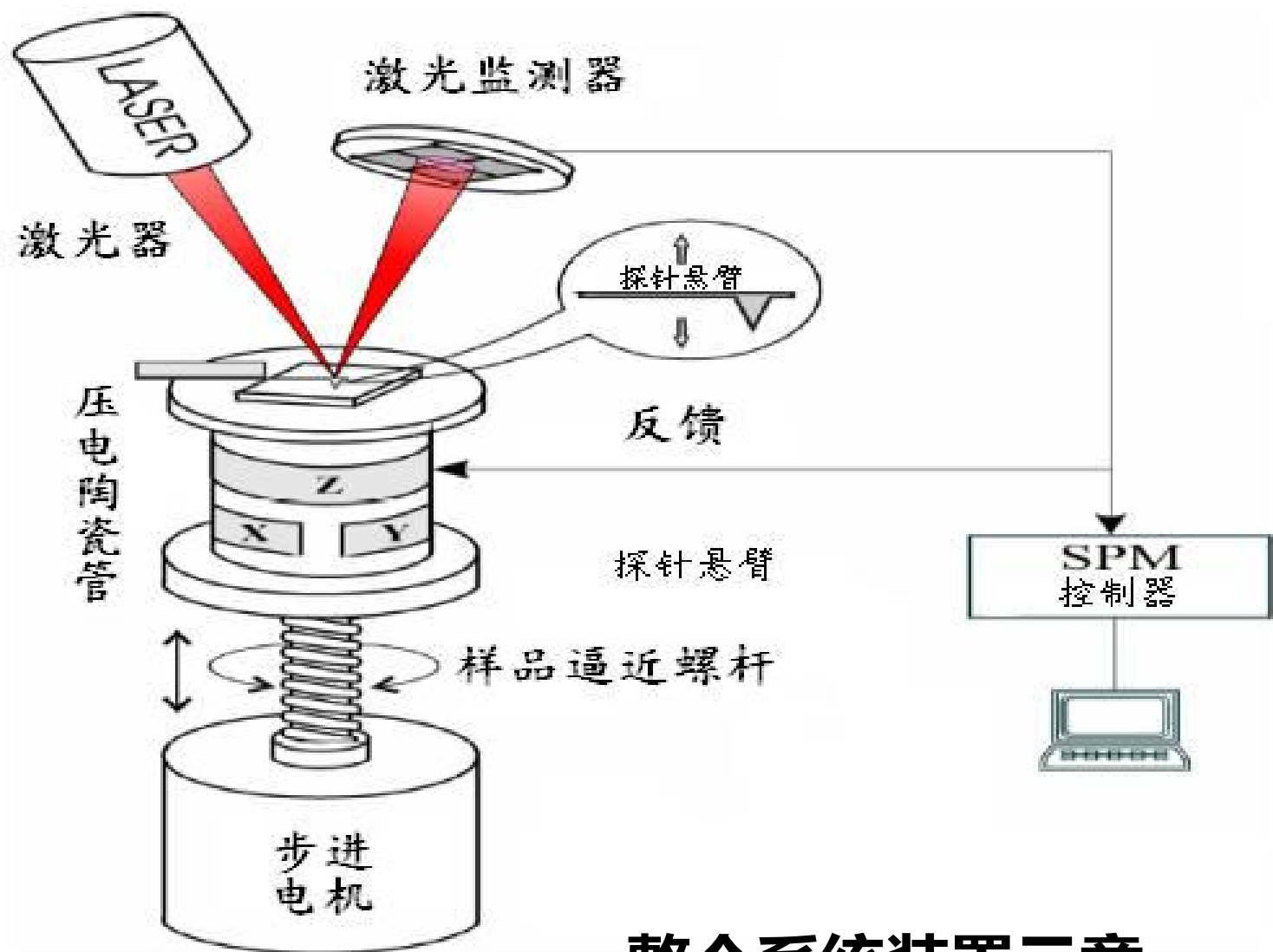
$$\Delta_i = \frac{\sqrt{2}d_{31}L^2}{\pi Dh}v_i \quad i = x, y,$$

$$\Delta_z = \frac{2d_{31}L}{h}v_z.$$

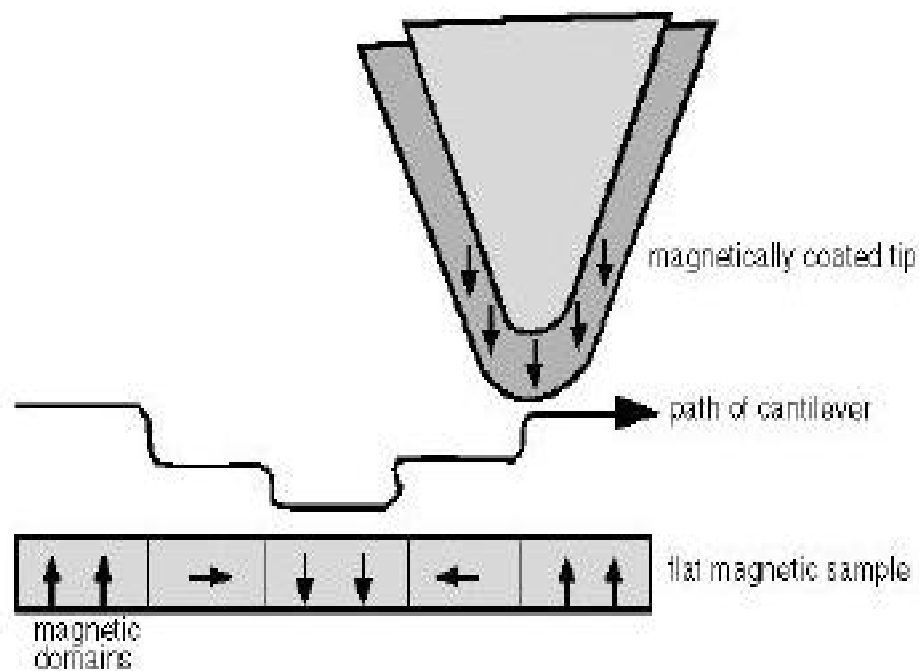
**L:** Tube length

**D:** Outside diameter

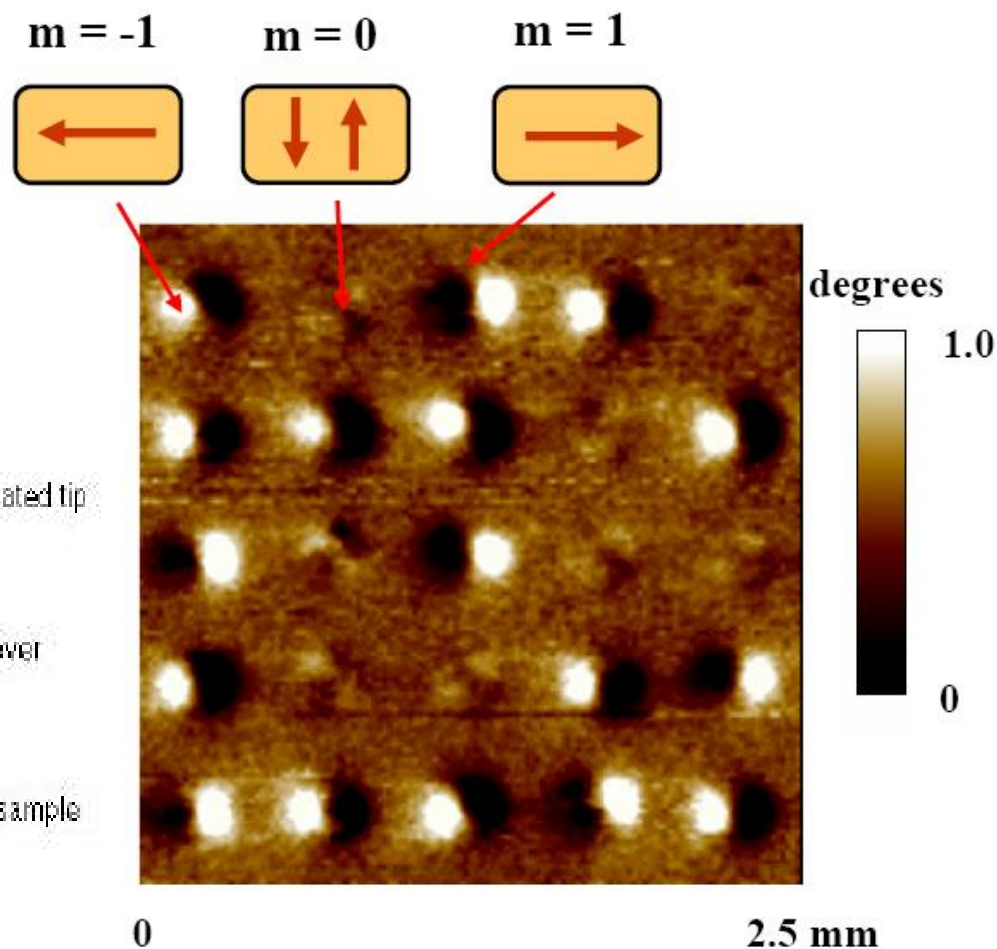
**h:** Tube thickness



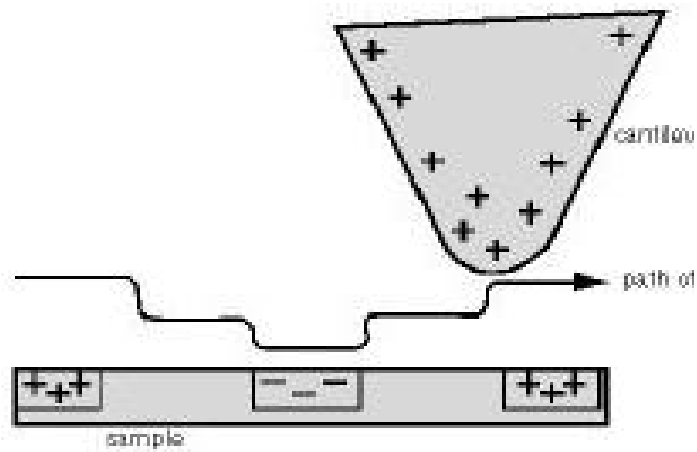
**整个系统装置示意**



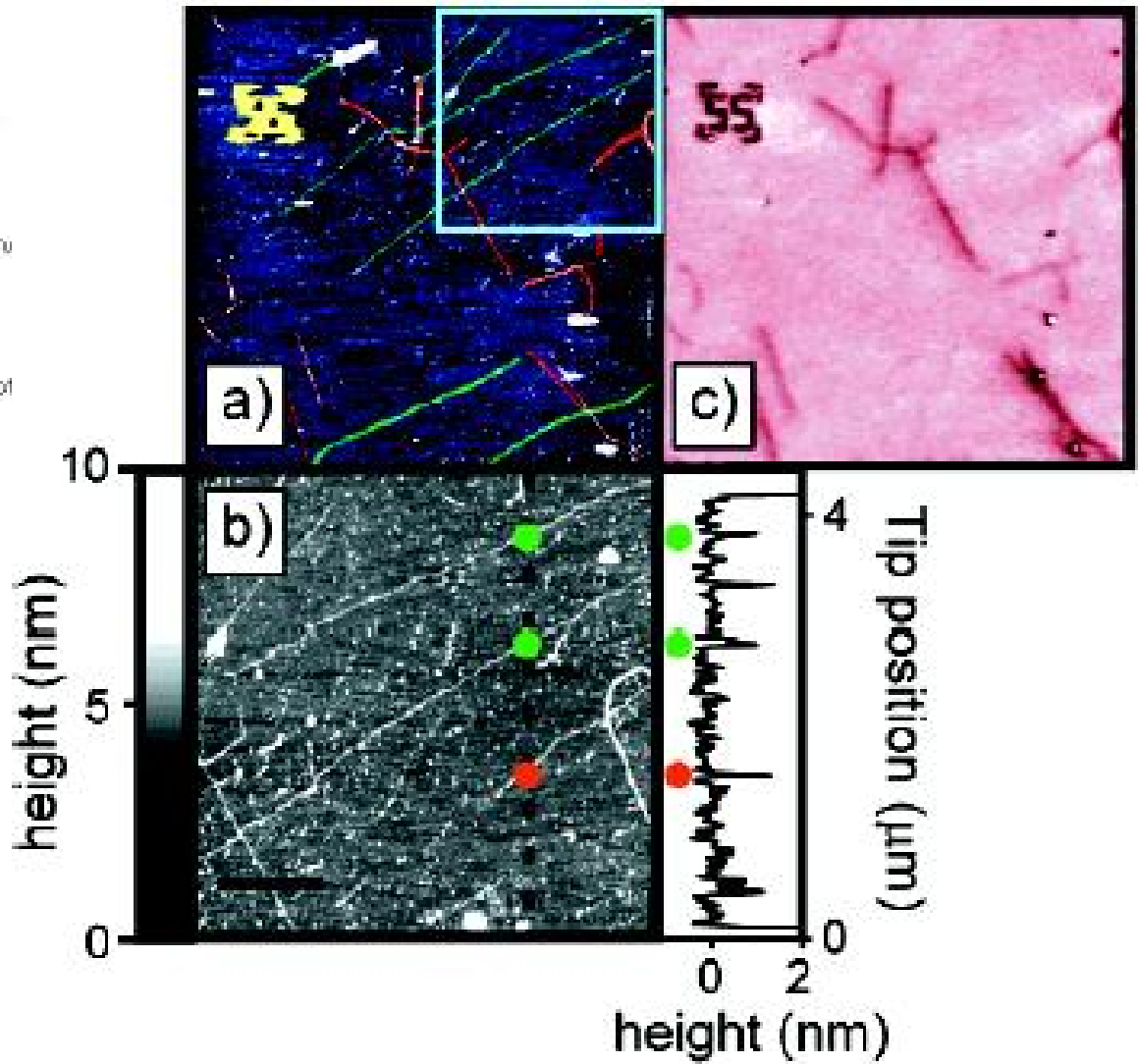
## MFM

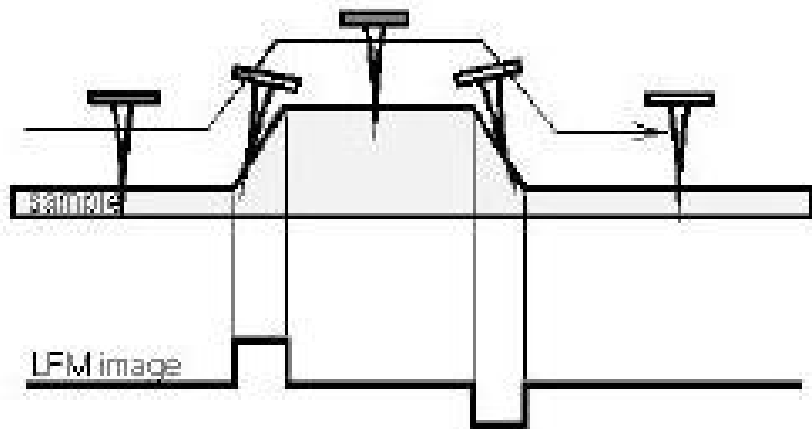
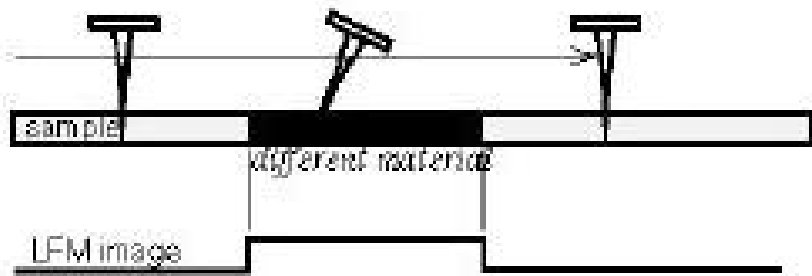


MFM image 100x140x20 nm Co particles  
(square lattice;  $s=300$  nm)

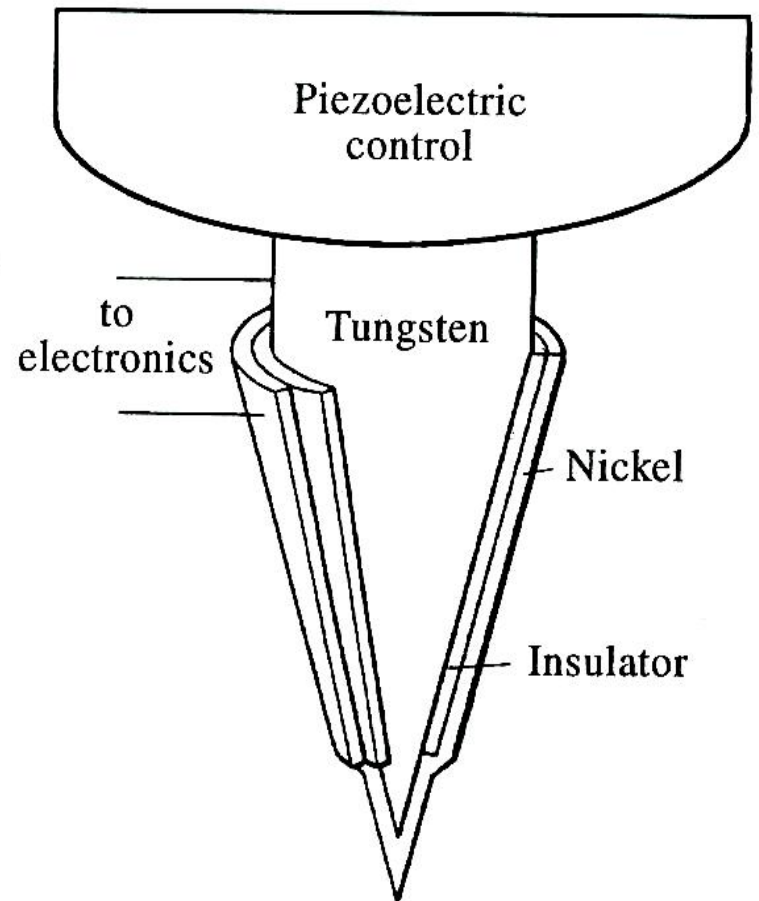


## EFM





**LFM**

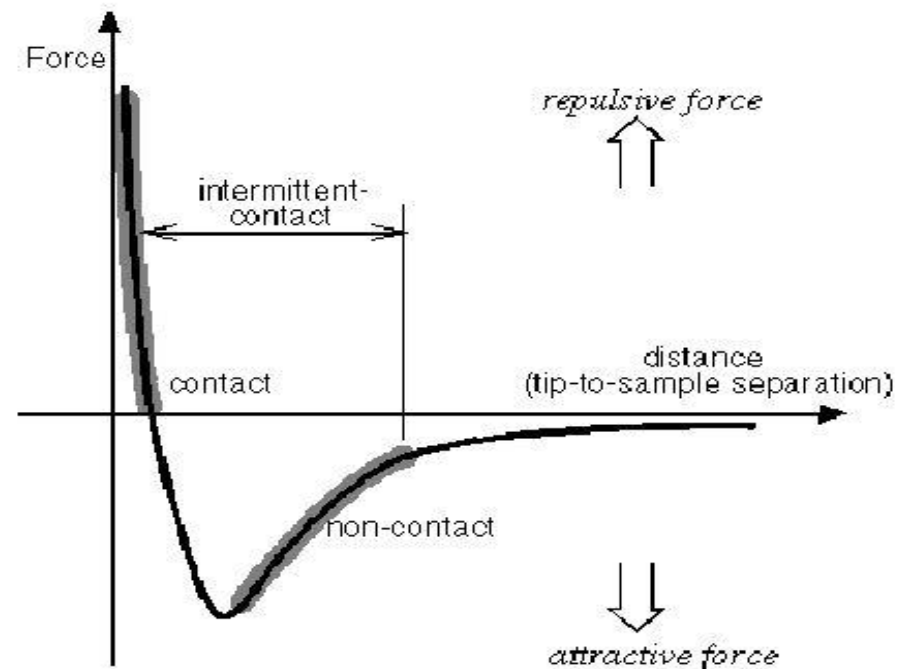


**SThM**

- 
- Scanning Tunneling Microscope (1981)
  - Scanning Near-Field Optical Microscope (1982)
  - Scanning Capacitance Microscope (1984)
  - Scanning Thermal Microscope (1986)
  - Atomic Force Microscope (1986)
  - Scanning Attractive Force Microscope (1987)
  - Magnetic Force Microscope (1987)
  - Frictional Force Microscope (1987)
  - Electrostatic Force Microscope (1987)
  - Inelastic Tunneling Microscope (1987)
  - Laser Driven STM (1987)
  - Ballistic Electron Emission Microscope (1988)
  - Inverse Photoemission Force Microscope (1988)
  - Near-Field Acoustic Microscope (1989)
  - Scanning Noise Microscope (1989)
  - Scanning Spin-precession Microscope (1989)
  - Scanning Ion Conductance Microscope (1989)
  - Scanning Electrochemical Microscope (1989)
  - Phonon Absorption Microscope (1989)
  - Scanning Chemical Potential Microscope (1990)
  - Kelvin Probe Microscope (1991)
  - Phase Mode (1995)
  - Magnetic acutated AC Imaging (1996)
  - Pulsed-Force Mode (1996)
  - Scanning Thermal Analysis - uTA release (1998)
  - Near Field Infra-red Microscopy (1999)
-

# 原子力显微镜工作模式

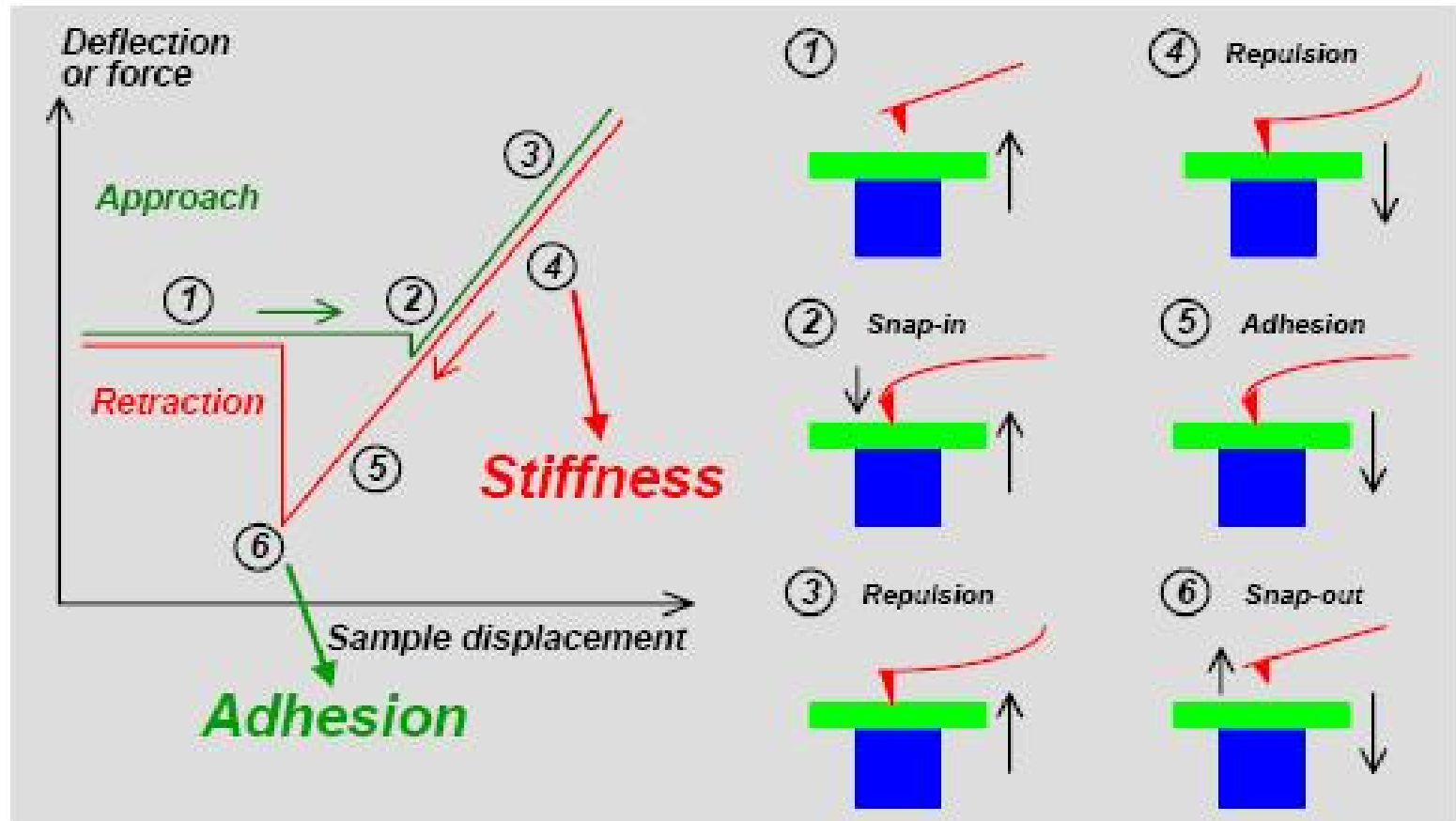
- Force-distance curve
- Contact mode
- Tapping mode
- Noncontact mode
- Others...

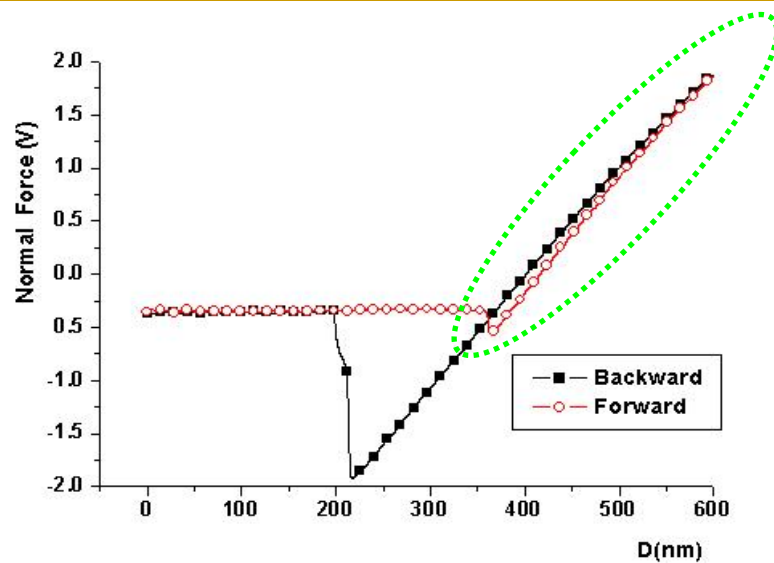


Interatomic force vs. distance curve.

# Force Curve

## Local measurement of tip-surface interactions:



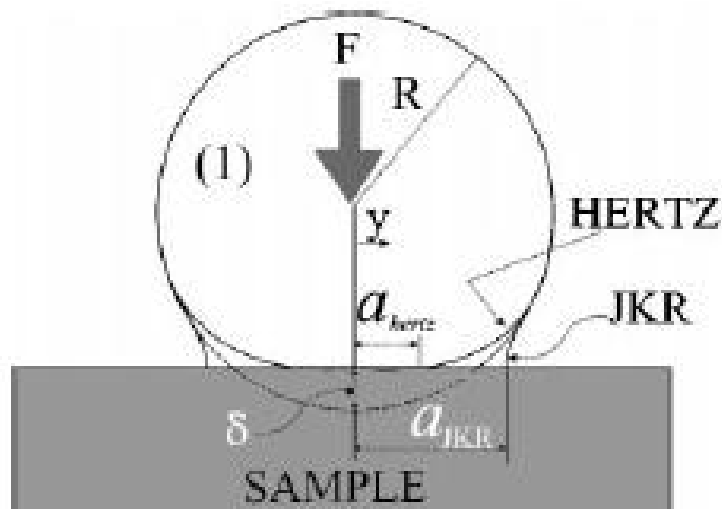


$$F_{\text{ad}} = 0,$$

$$a = \sqrt[3]{\frac{RF}{K}},$$

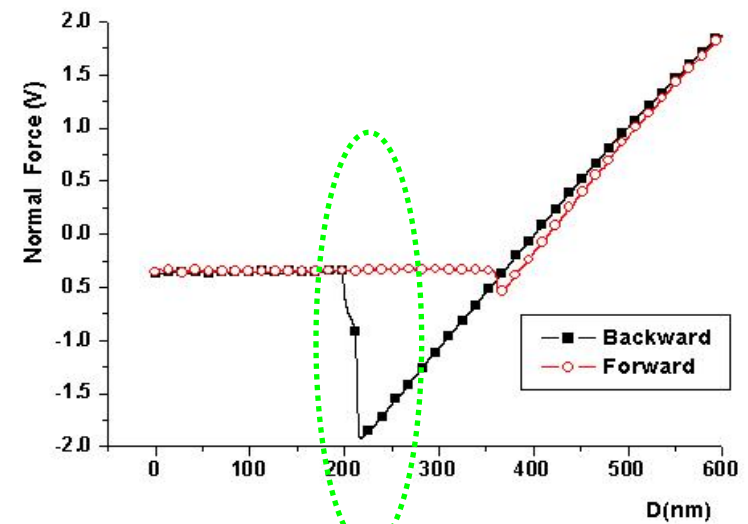
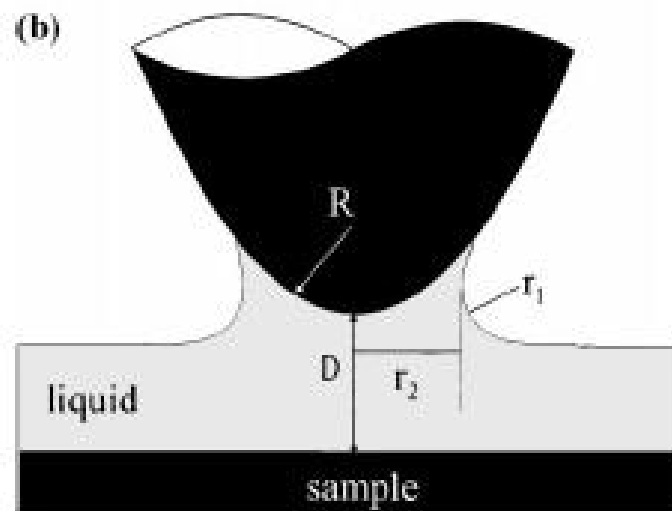
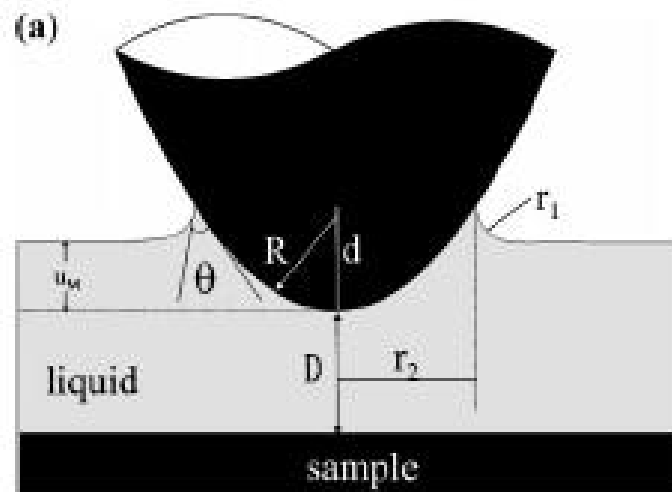
$$a_0 = 0,$$

$$\delta = \frac{a^2}{R} = \frac{F}{Ka},$$



**Contact**

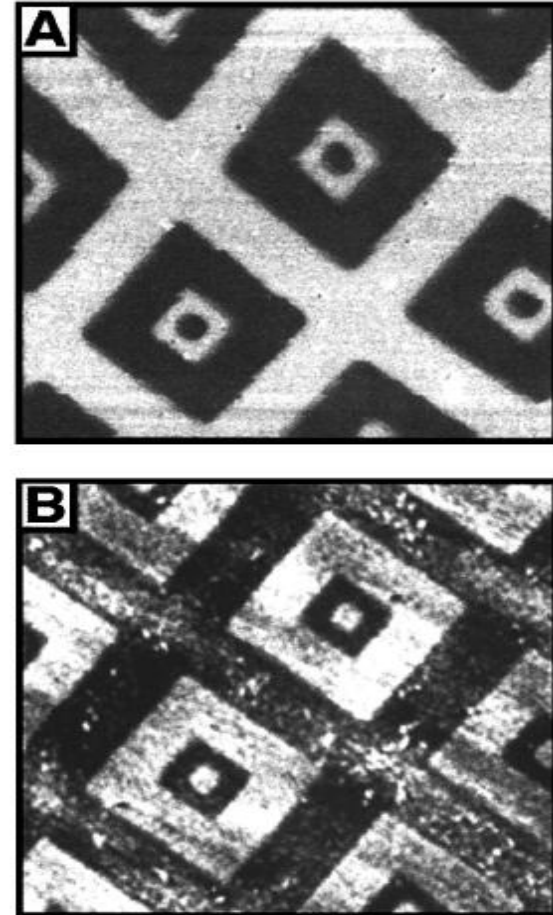
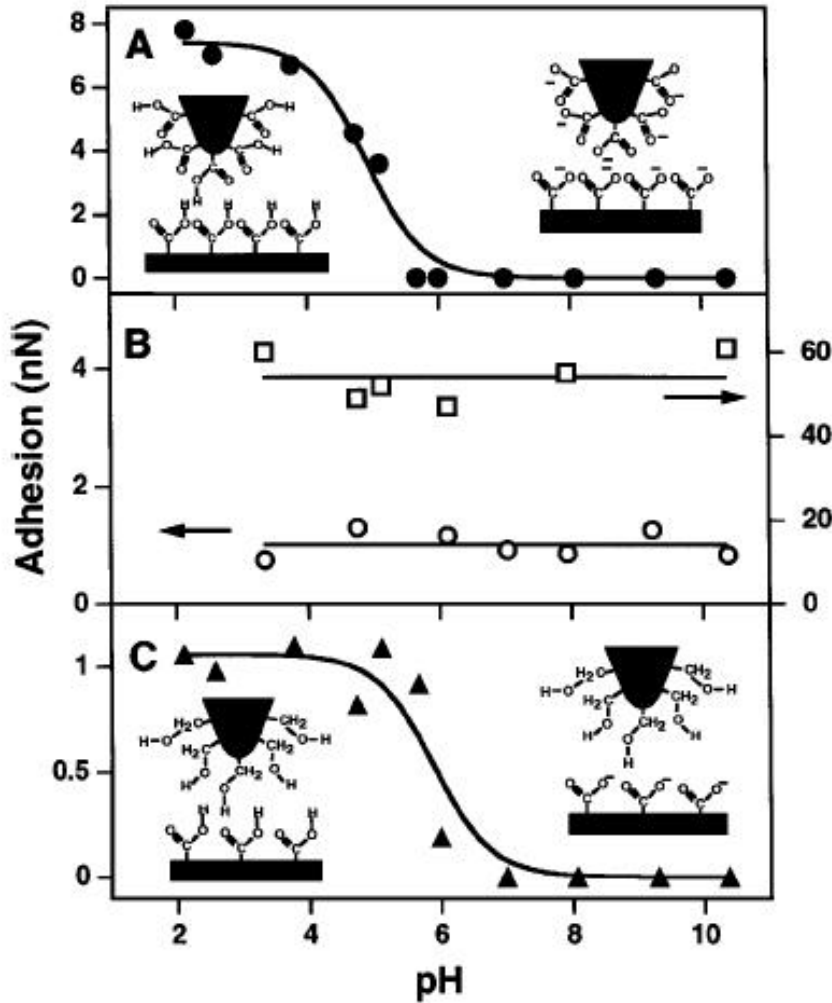
$$\frac{1}{K} = \frac{3}{4} \left( \frac{1 - \nu^2}{E} + \frac{1 - \nu_i^2}{E_i} \right)$$



$$F = - \frac{4\pi R \gamma_L \cos \theta}{[1 + D \log(H_r)] / 1.08 \cos \theta},$$

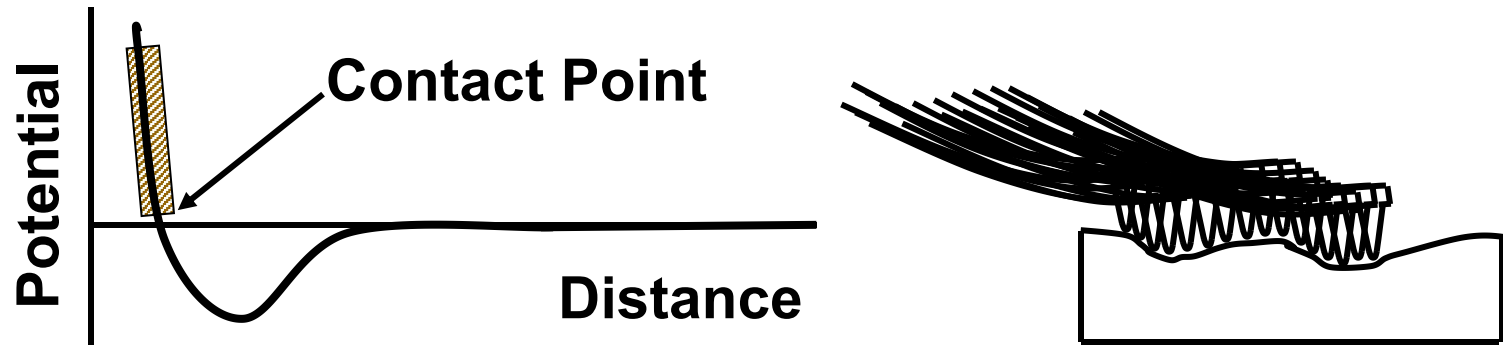
Adhesion

表面亲、疏水性能



**Chemical force**  $-\text{CH}_3, -\text{COOH}$

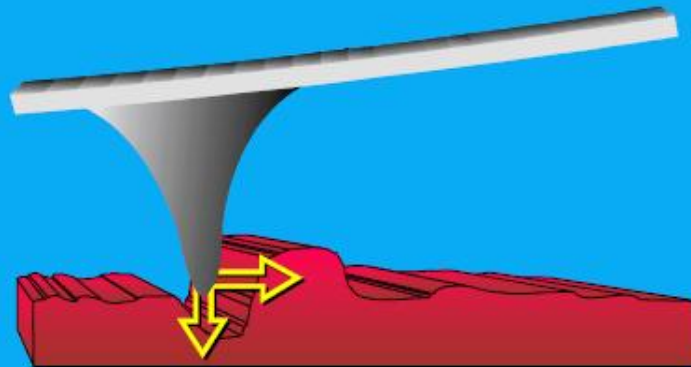
# Contact mode



DC Mode:

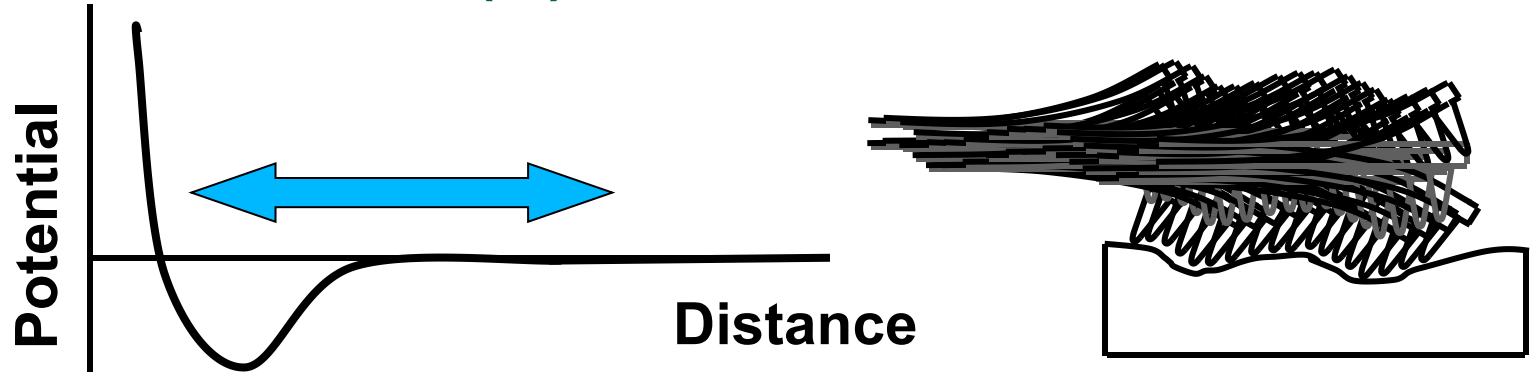
Typical imaging force:  
 $10^{-7} - 10^{-9} \text{ N}$

**Contact AFM**  
-Easier setup



Soft Sample

# Dynamic mode(s)

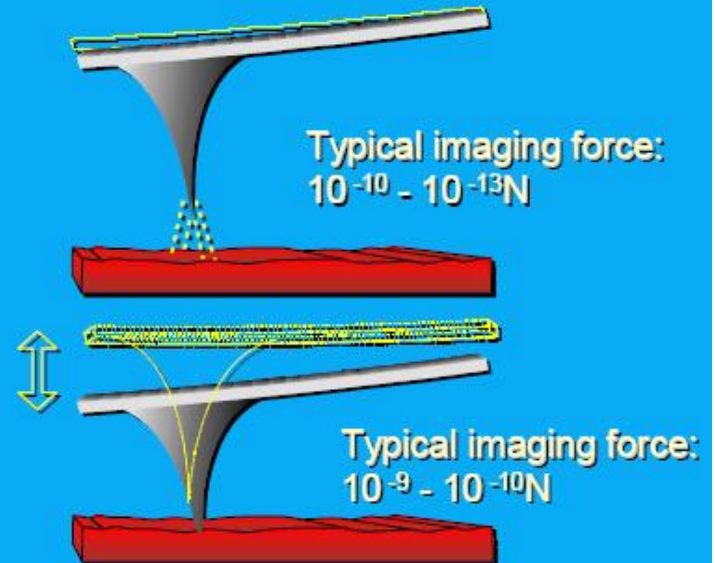


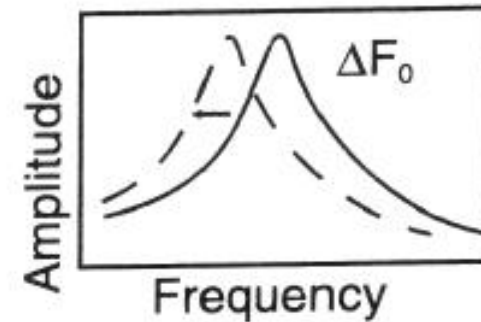
## Non-contact AFM

-Does not damage sample

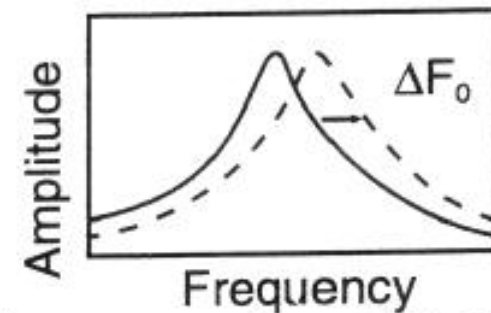
## Intermittent-contact AFM

-Good for soft samples





Attractive gradient equivalent to additional spring in tension attached to tip, reducing the cantilever resonance frequency.



Repulsive gradient equivalent to additional spring in compression attached to tip, increasing the cantilever resonance frequency.

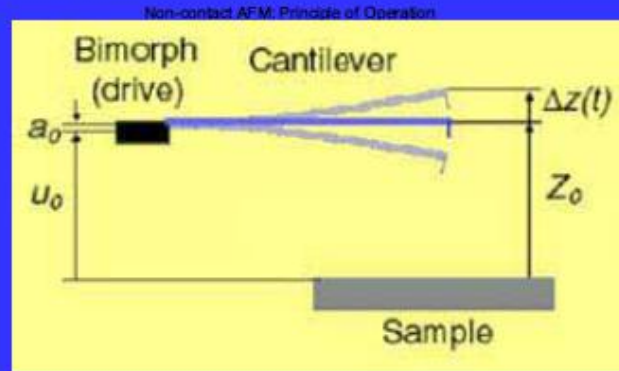
Effective resonant frequency: 
$$\omega_e = \sqrt{\frac{k_c - dF_{ts} / dz}{m}}$$

# Non-contact mode

## Non-contact AFM

$$u(t) = u_0 + a_0 \exp(i\omega t)$$

$$\tau = 2Q / \omega$$



$$z(t) = z_0 + \Delta z(t)$$

$$m_c \ddot{z}(t) + \gamma_c \dot{z}(t) + k_c [z(t) - a_0 \exp(i\omega t)] = F_{ext}$$

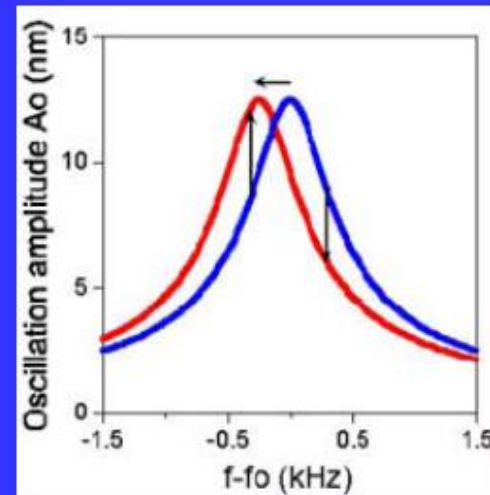
$$\omega_0' = c (k_c - dF/dz)^{1/2}$$

$k_c$  - cantilever spring constant

$c$  - function of the cantilever mass

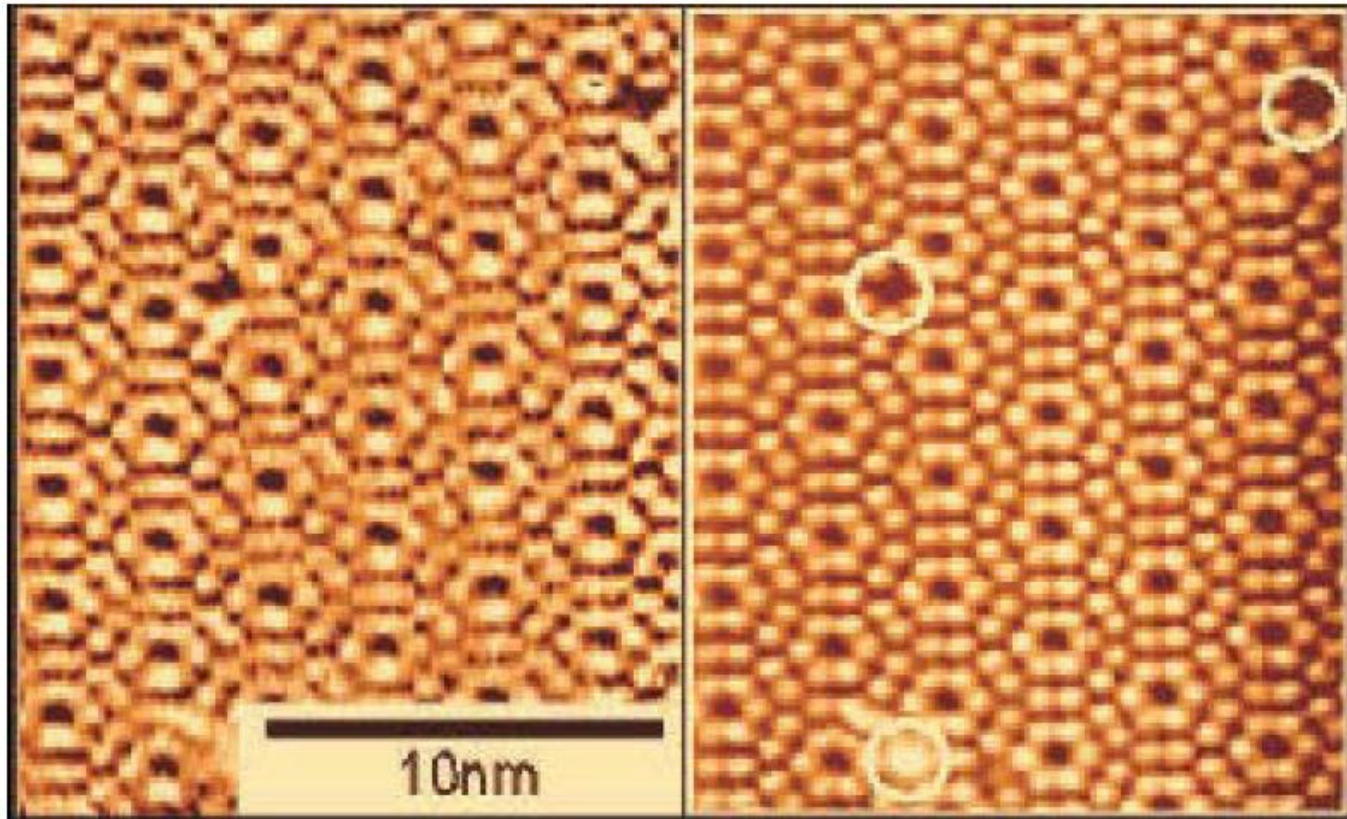
$A_0$  - amplitude at resonance

$Q$  - quality factor



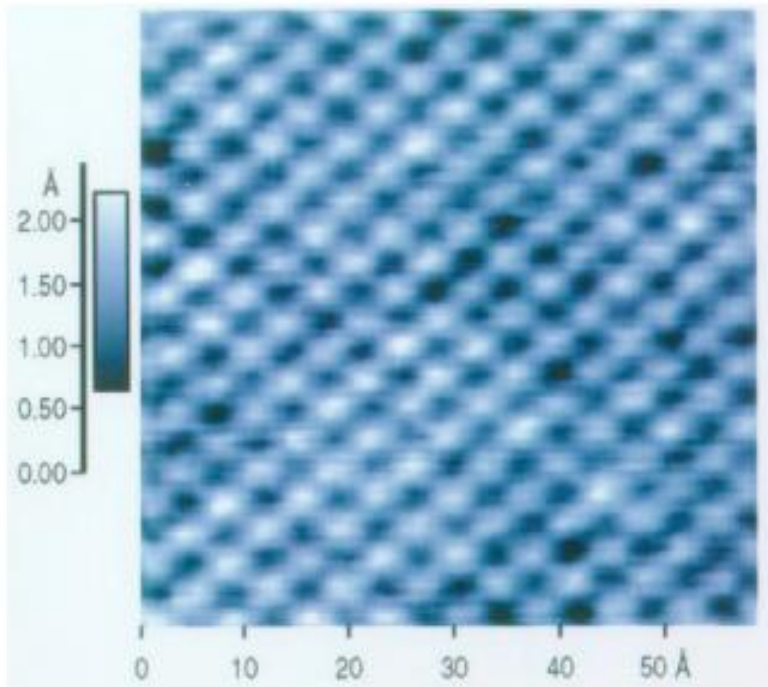
(or Frequency-modulation AFM)

## Si(111)-(7x7) AFM vs. STM

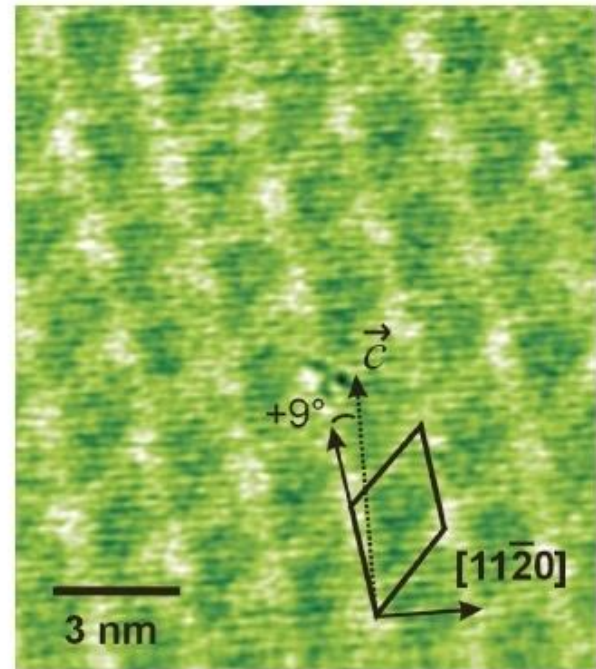


AFM

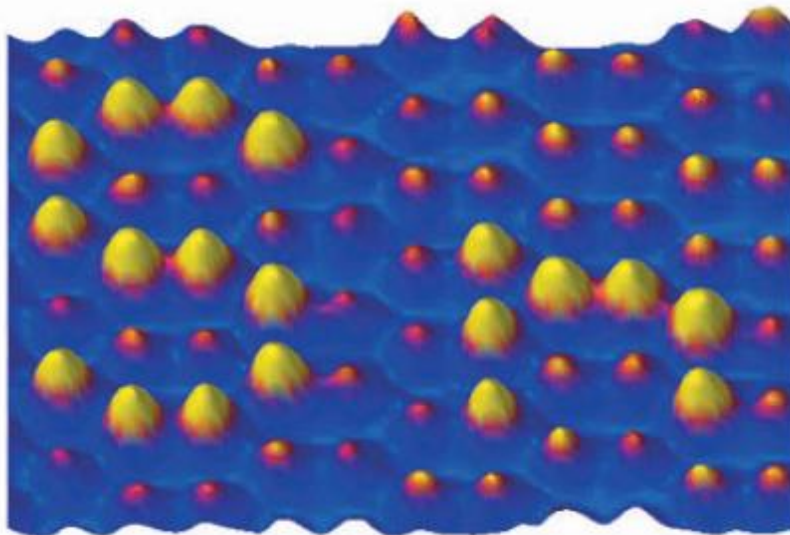
STM



**KCl**



**Al<sub>2</sub>O<sub>3</sub>**



**Sn on Ge(111)**

**Atom inlay**

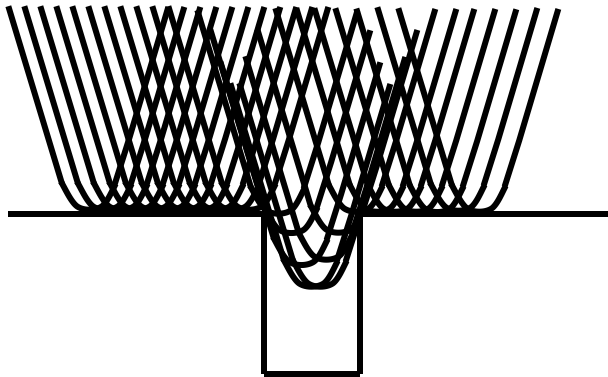
# 测量误差分析

## Error sources:

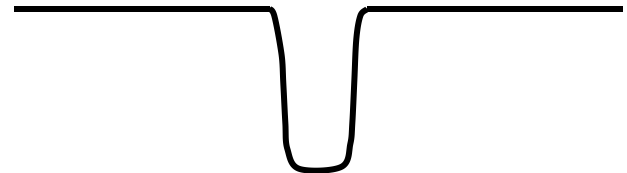
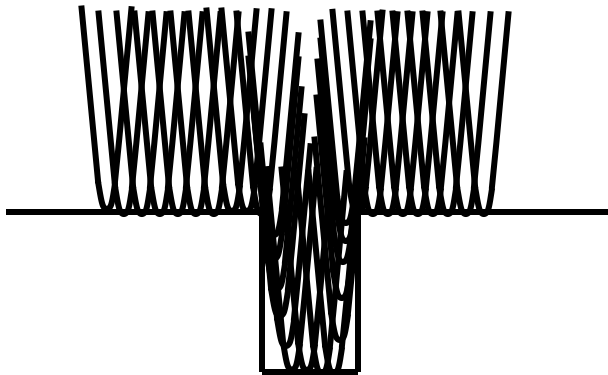
- Tip effect
- Piezo-scanner
- Mechanical vibration
- Electronic fluctuation
- D/A, A/D conversion error
- Environmental effect
- .....

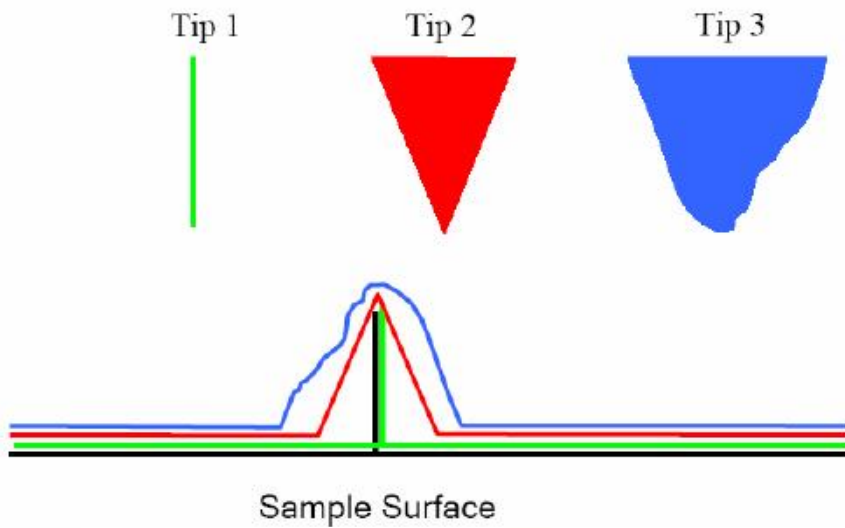
# Tip effect

**Dull Probe**

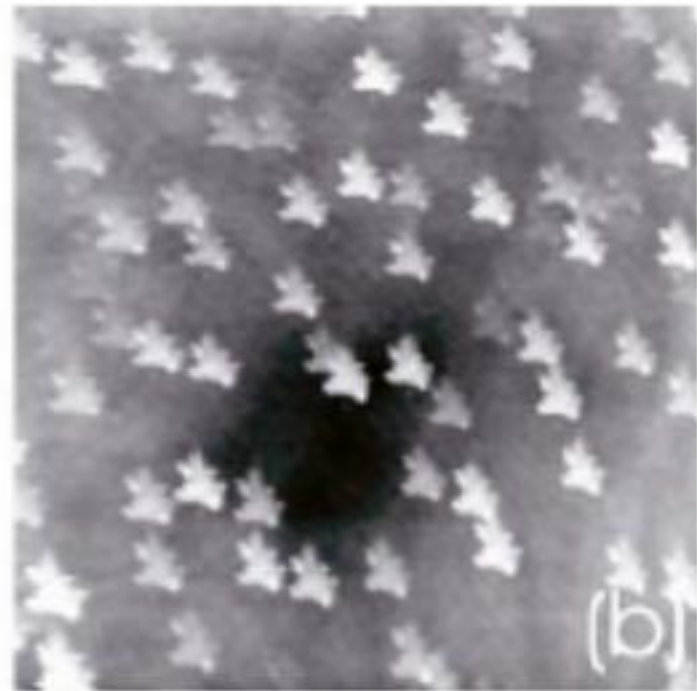
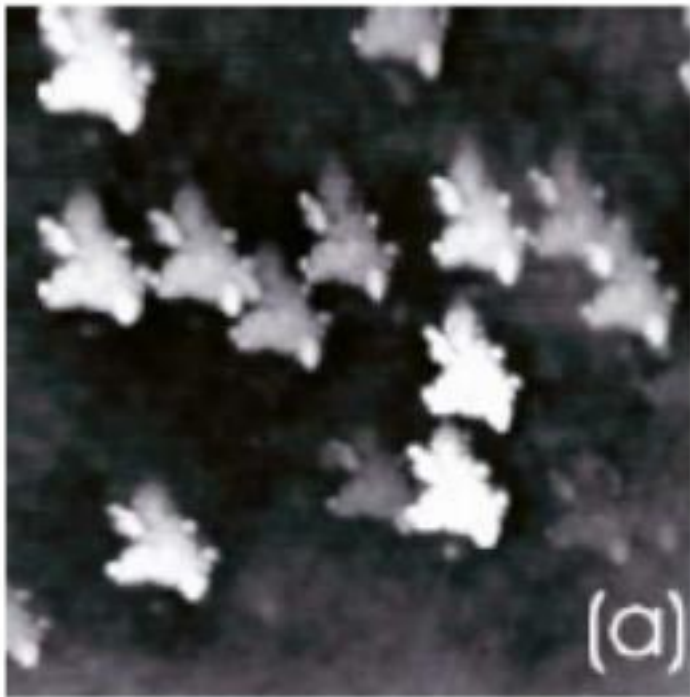


**Sharp Probe**

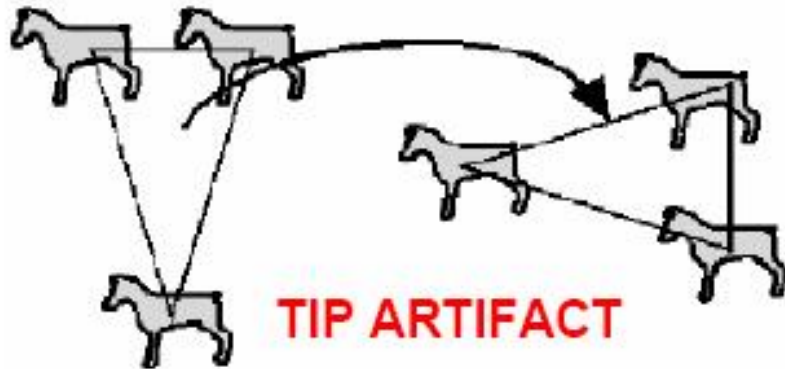




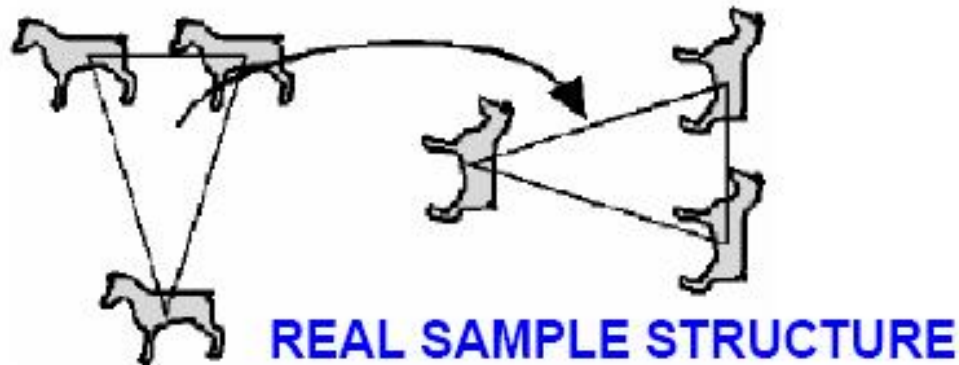
**AFM image of a sample with  
needle-like structures**



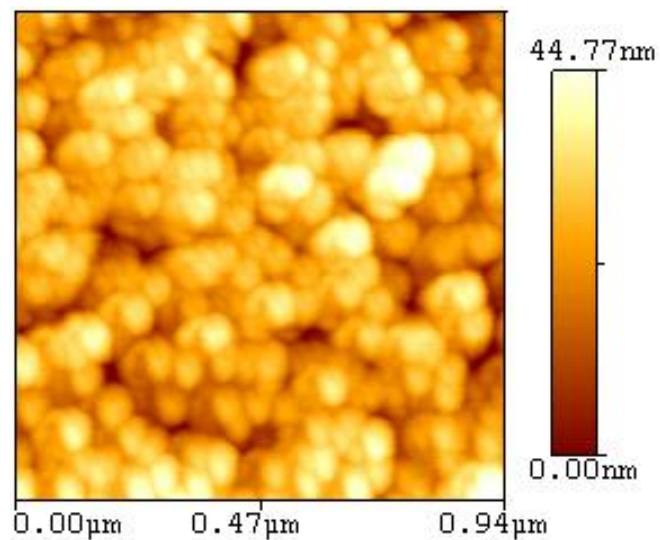
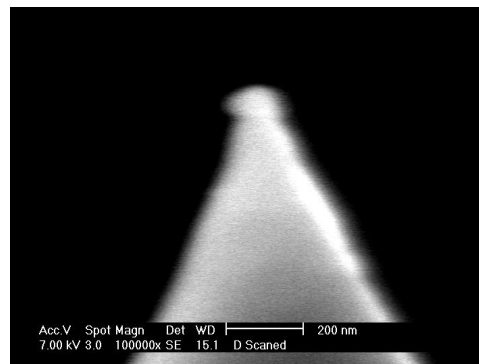
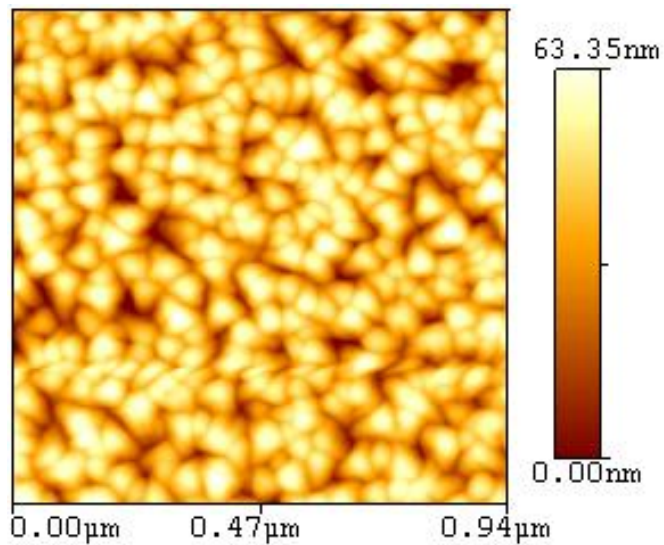
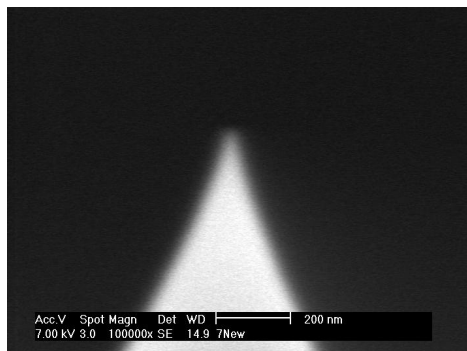
# How to identify tip-induced artifacts?



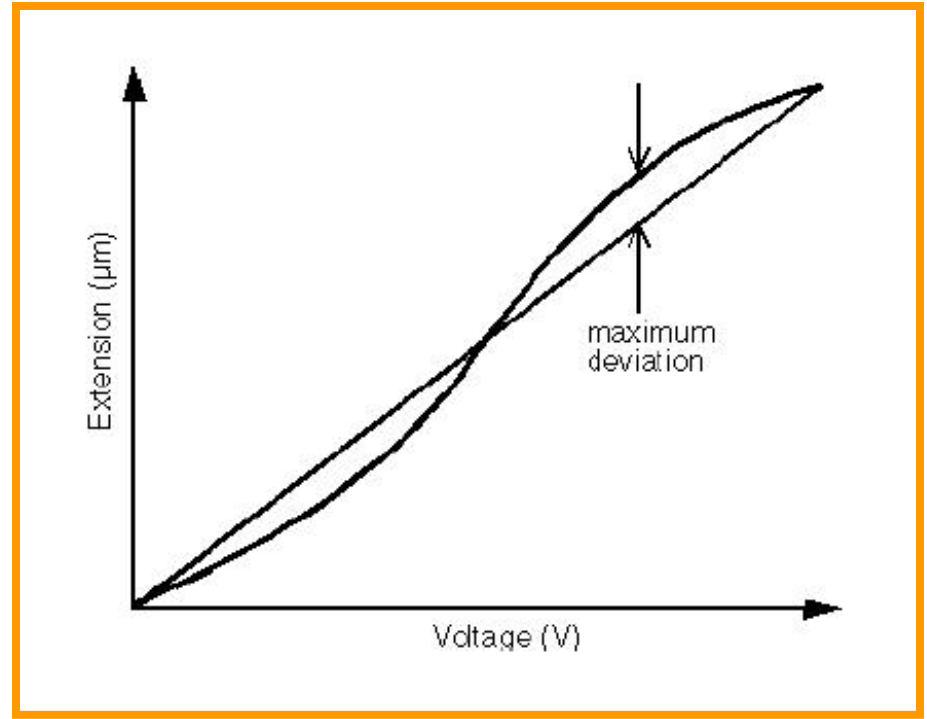
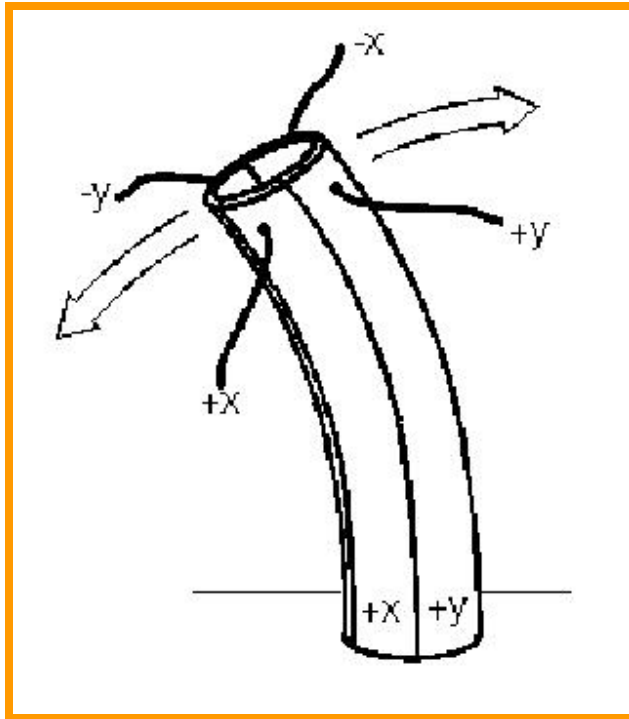
Features of the image **DO NOT rotate** → tip artifact!



Features of the image **DO rotate** → real sample structure!



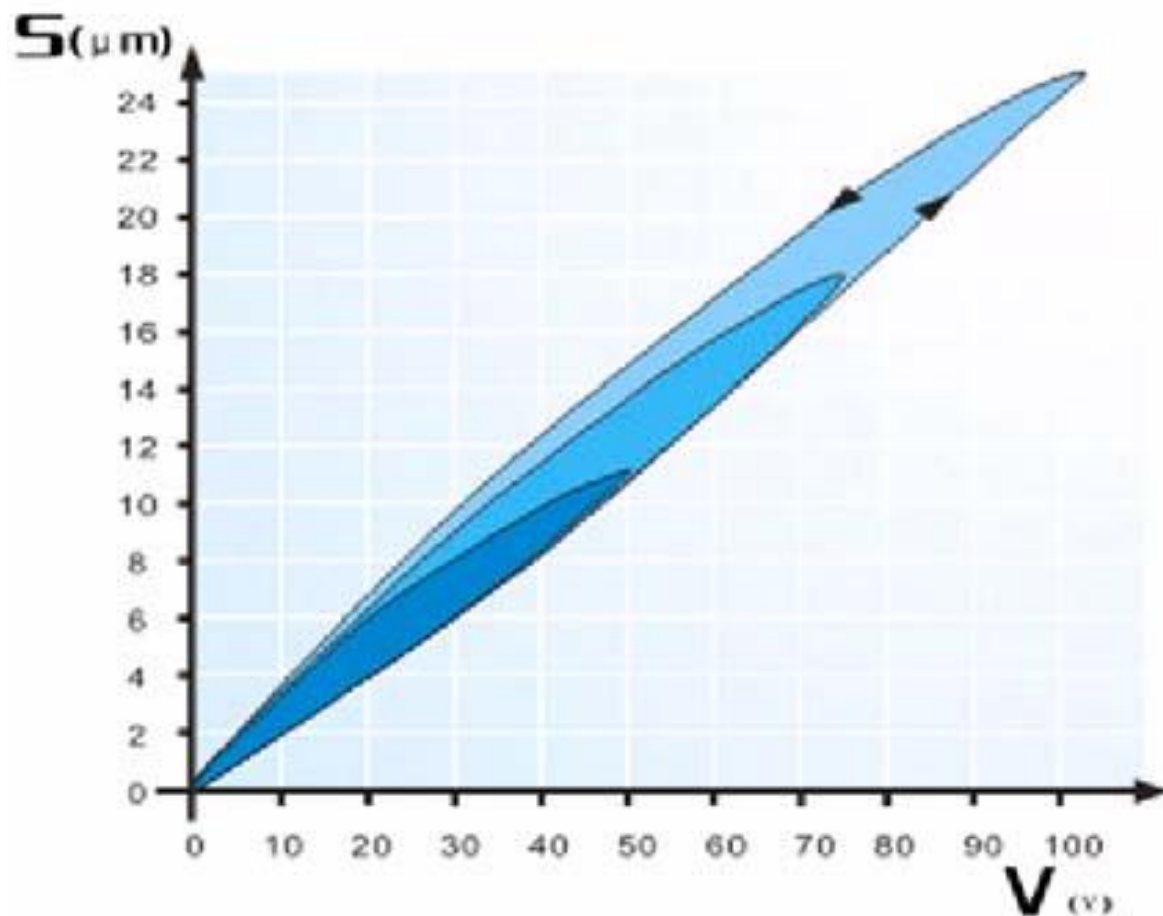
**Good/Bad Tip**

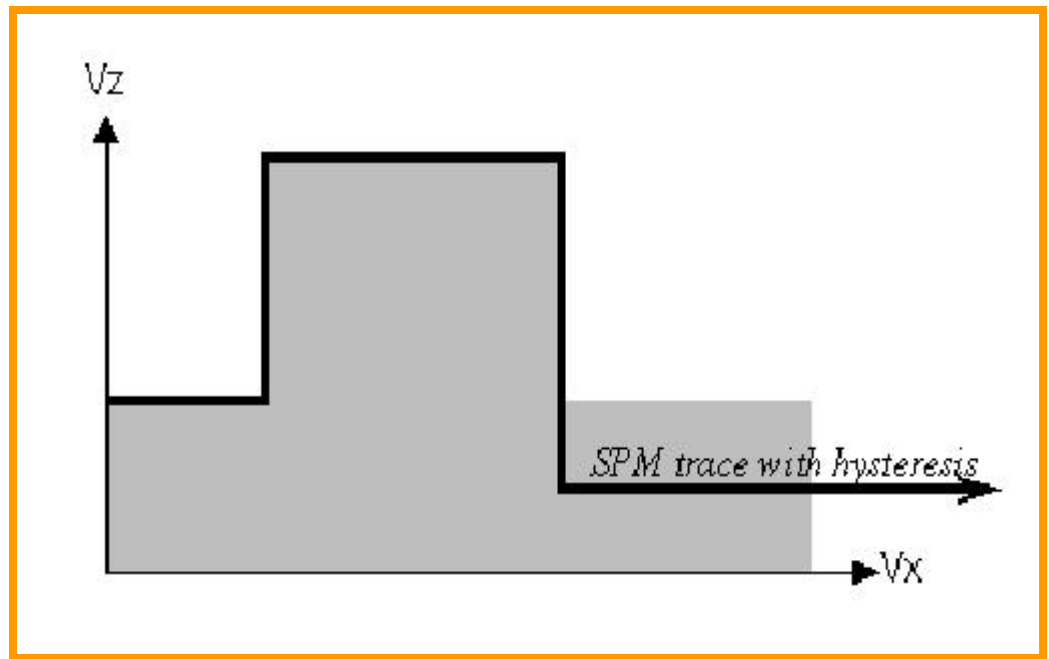
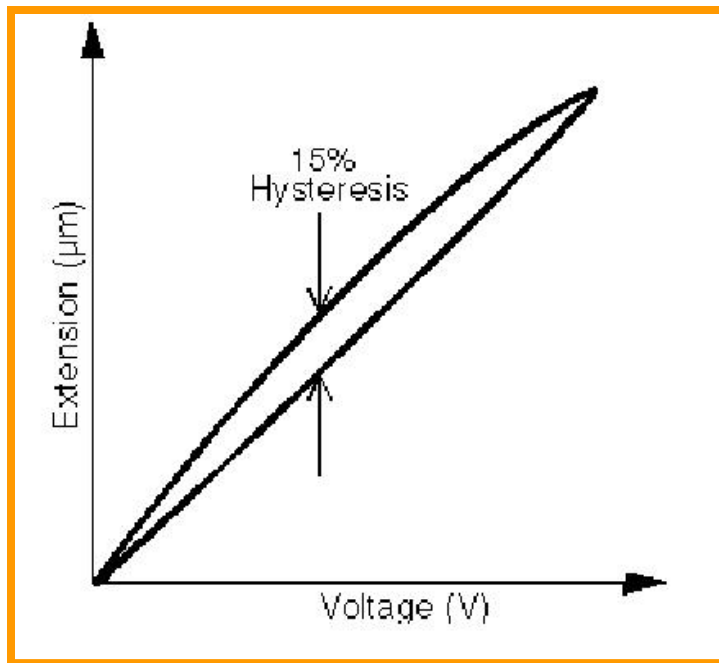


**Piezo scanner and its intrinsic nonlinearity.**  
**Nonlinearity: typically 2%-25%**

**迟滞特性** • 压电陶瓷的升压和降压曲线之间存在位移差称为迟滞现象。

(主要因素：压电陶瓷分子间存在阻力和残余极化)





**Hysteresis and its effect on a step.**

**Hysteresis: typically 15%-40%**

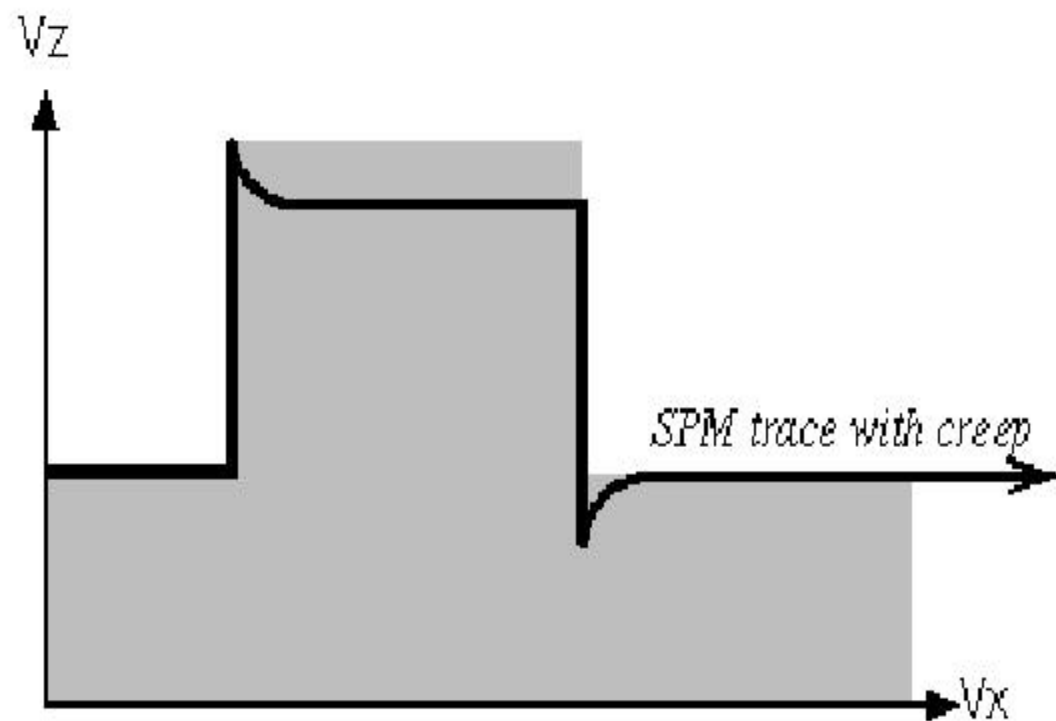
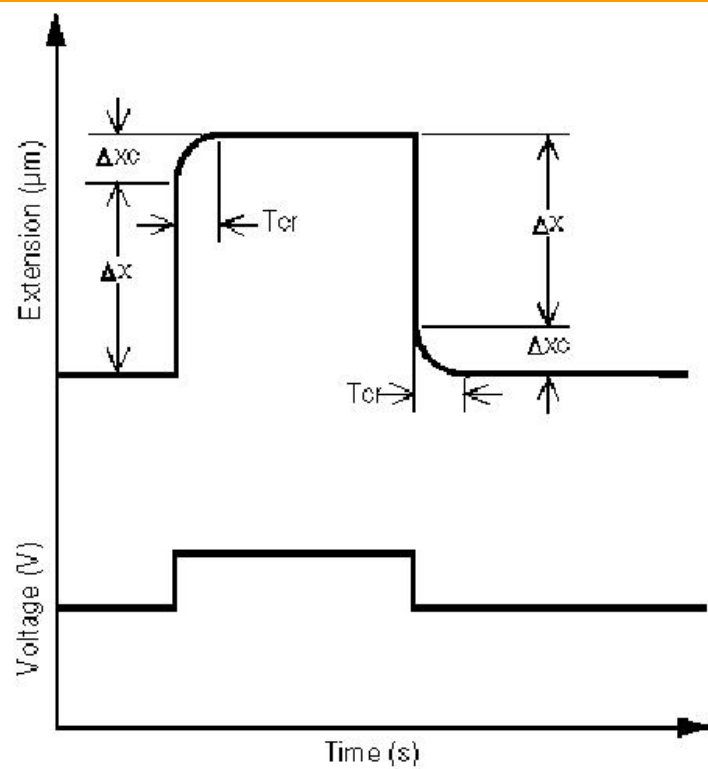


**Scan direction: left→ right**

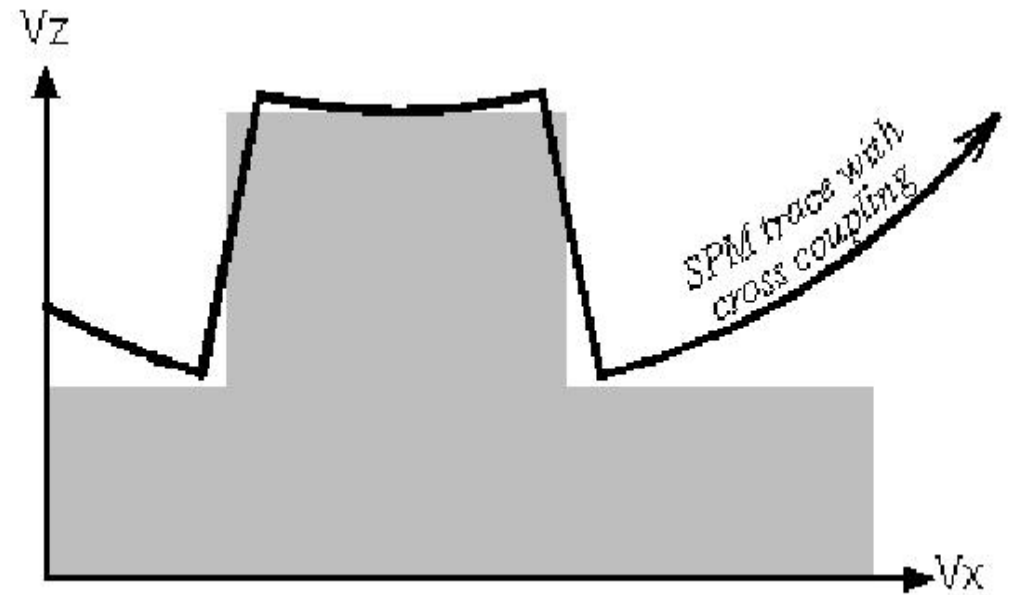
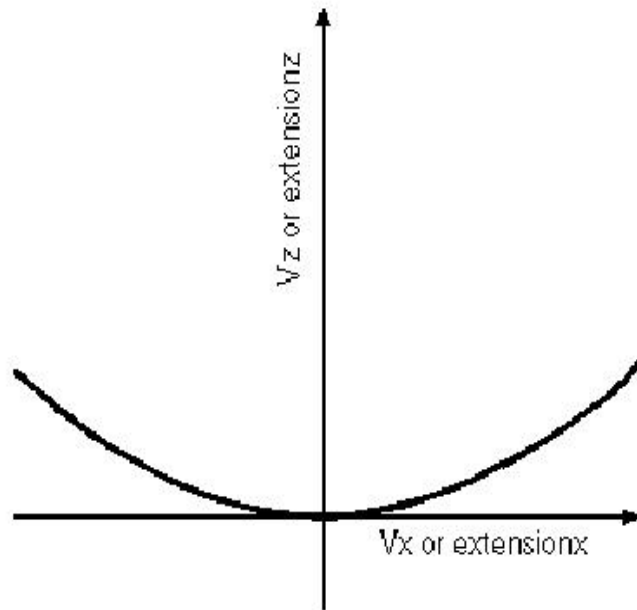


**Scan direction: right→ left**

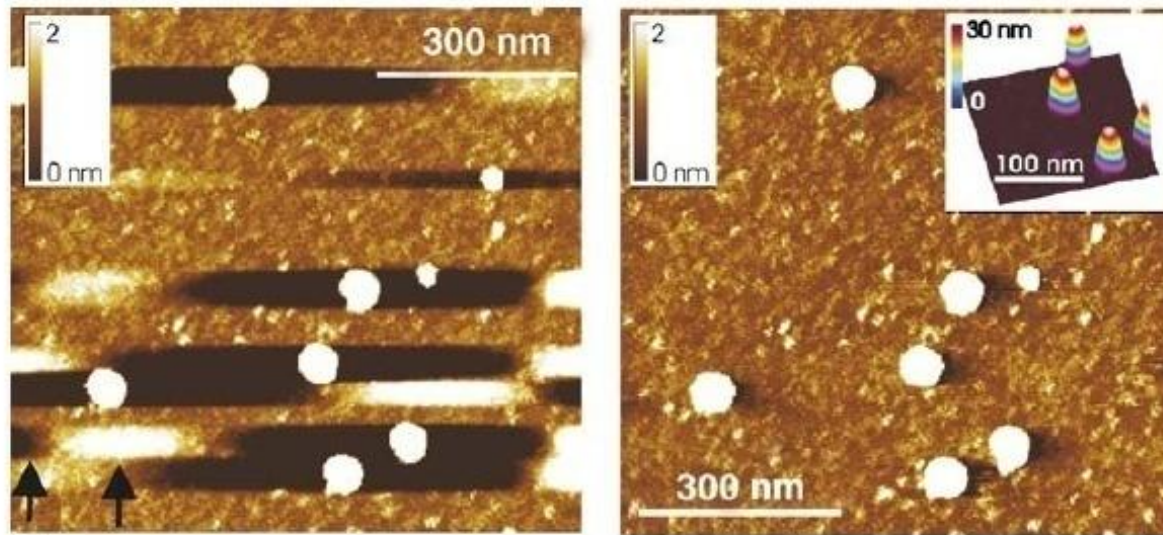
**Grating period:  $3\mu\text{m}$**



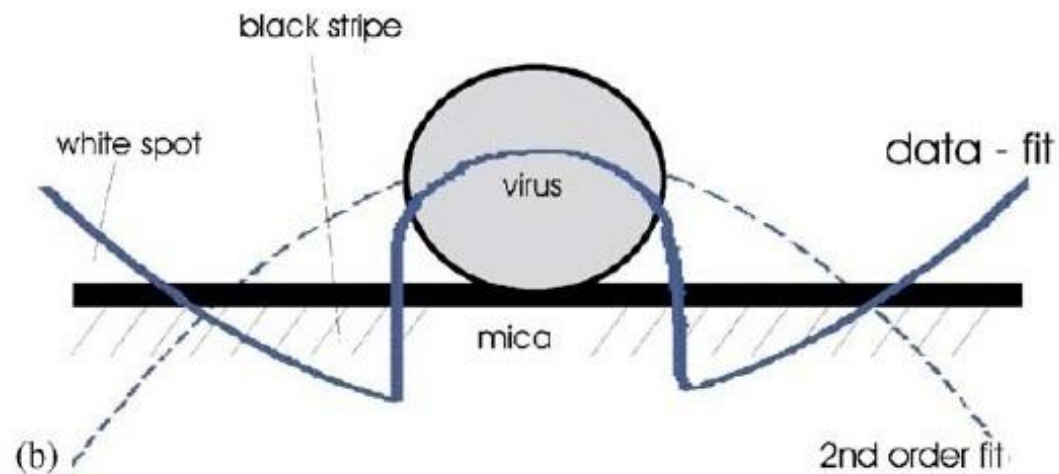
**Creep and its effect on a step.**



**Cross coupling and its effect on a step.**

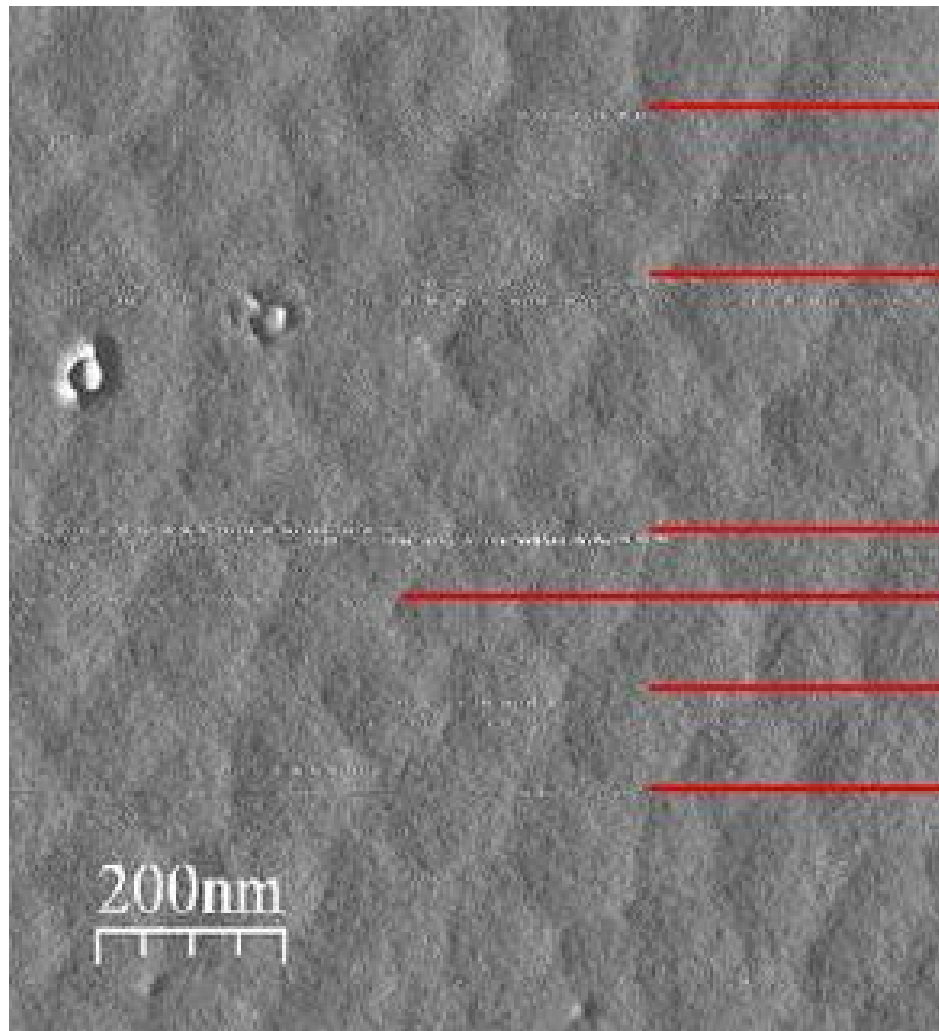


(a)



(b)

## Image processing artifacts



knocking on hood

laughing

door slamming

jumping on the floor

clapping

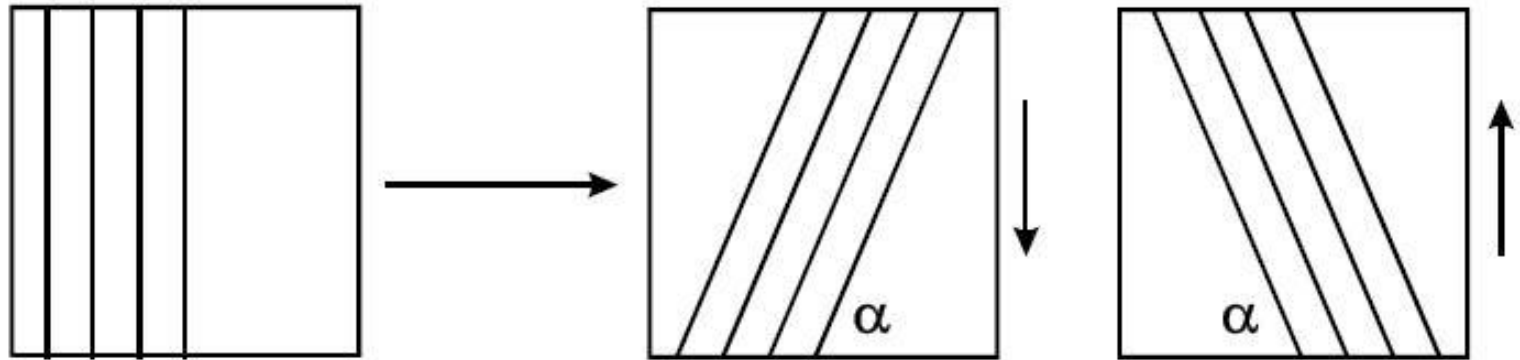
kicking the table

**Vibrations ( noises)**

## thermal drifts:

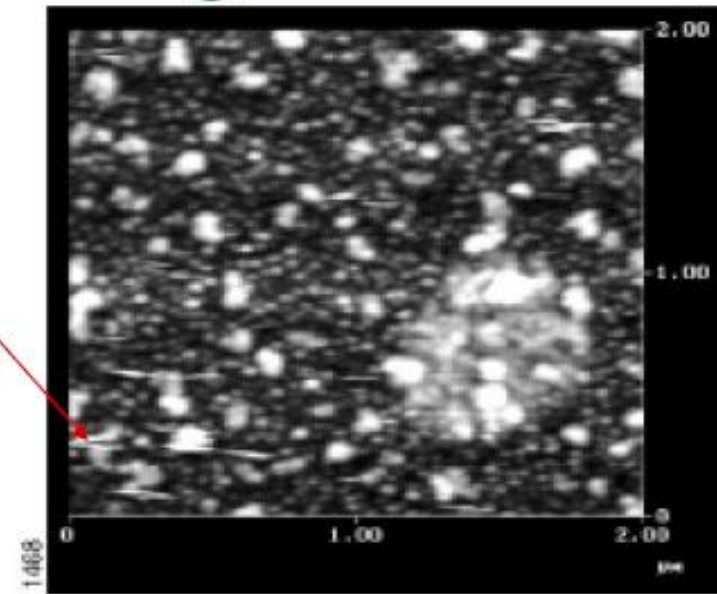
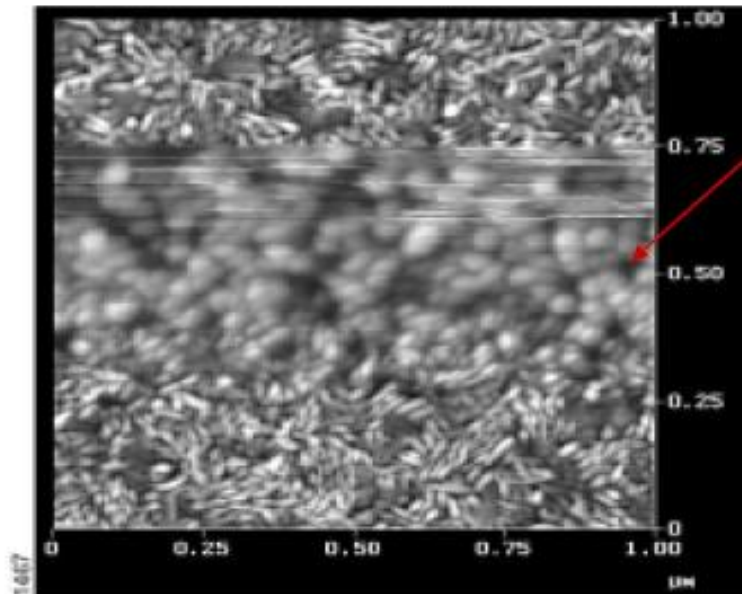
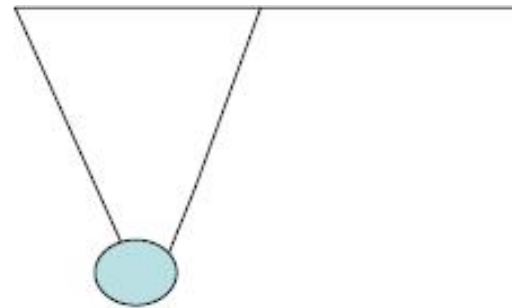
e.g.

Y – scan disabled

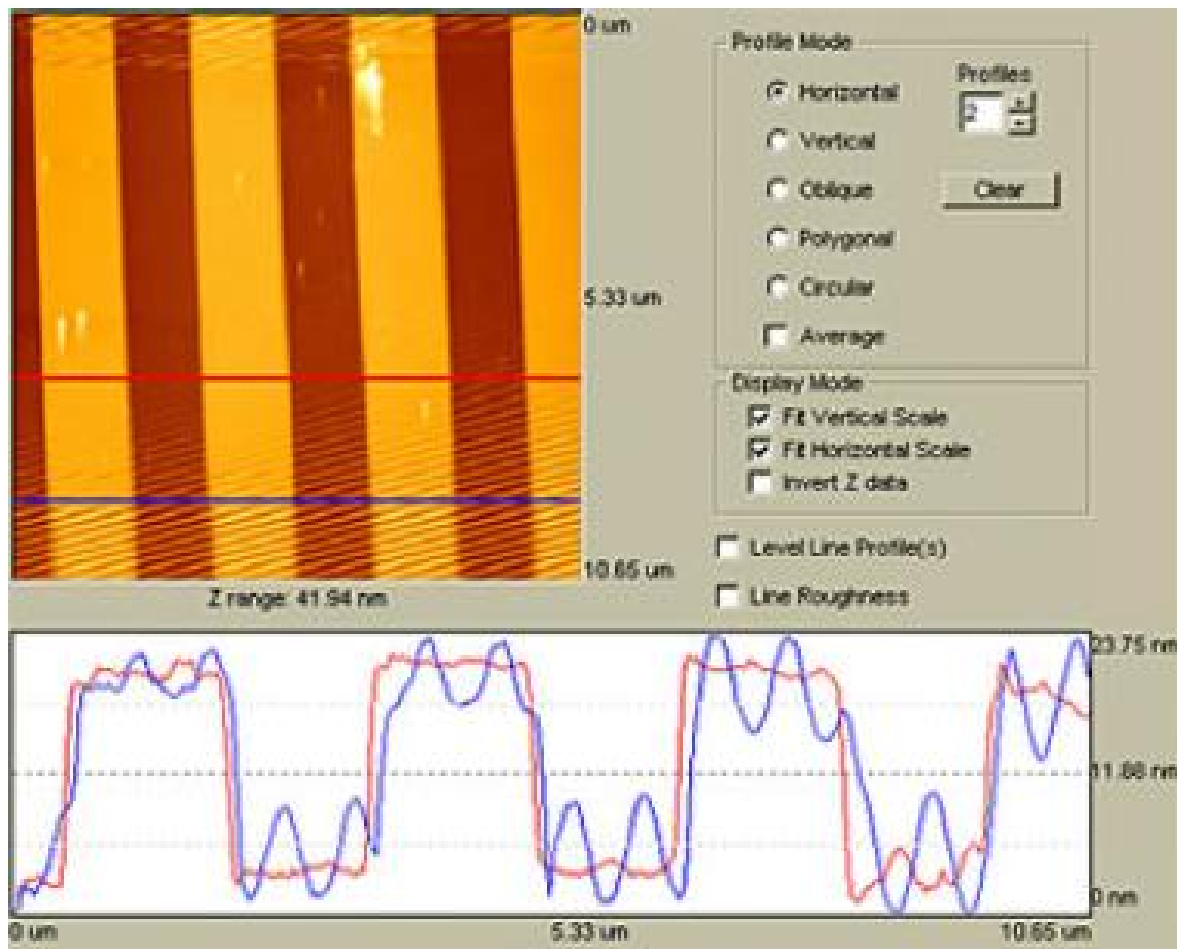


→ wait for stabilization

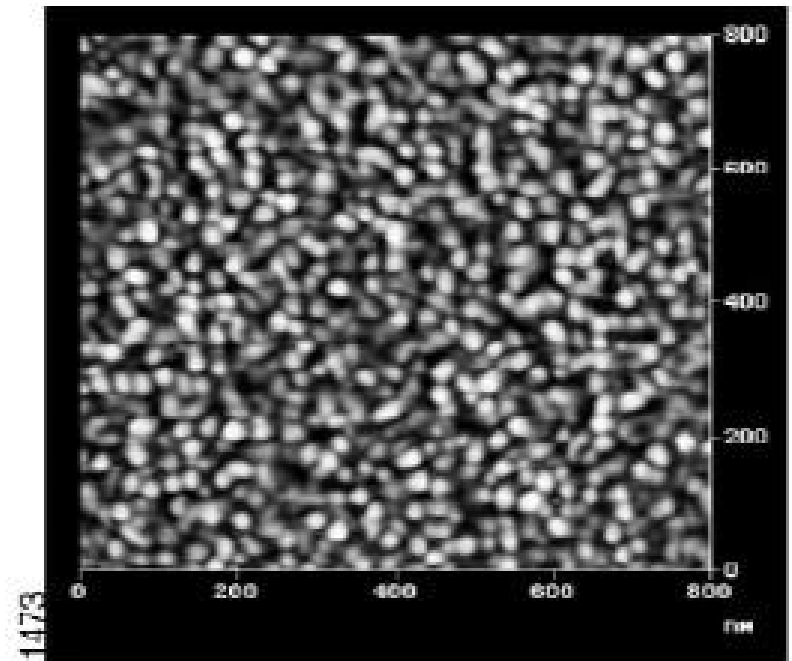
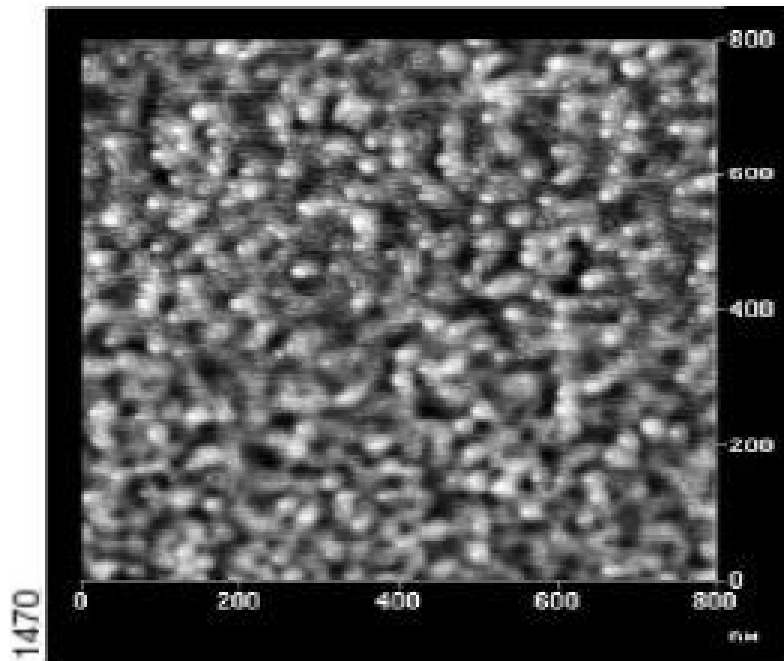
Blurring or Streaking that comes and goes



**Contamination**



**Electronic noises**



**Feedback parameters**